



Corrosion Mitigation Of Copper In Acidic Solutions Using N-(Pyridin-2-Ylmethylene) Aniline: Electrochemical, Adsorption And Thermodynamic Studies

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ABSTRACT

The synthesized Schiff base N-(pyridin-2-ylmethylene)aniline was employed as a corrosion inhibitor and it was studied for the inhibition performance for copper in 1.0 M HNO₃ and 0.5 M H₂SO₄ solutions using gravimetric, electrochemical, adsorption and thermodynamic techniques. The Schiff base was prepared from the condensation of pyridine-2-carboxaldehyde with aniline. The formation of the azomethine (C=N) was confirmed by FTIR. The results of weight-loss measurements showed that the inhibition efficiency increased with the concentration of inhibitor in the medium, reaching 77.5% in 1.0 M HNO₃ at 1.00 mM and 93.7% in 0.5 M H₂SO₄. The results show that the inhibitor is better in sulphuric acid. Analysis using electrochemical impedance spectroscopy illustrated a notable surge in charge-transfer resistance. The double-layer capacitance and constant phase element values were found to decrease. This observation reveals the formation of a compact protective film on the copper surface. Potentiodynamic polarization studies have indicated that the Schiff base causes a significant decrease in the corrosion current density with a small shift in the corrosion potential suggesting it to be a mixed-type inhibitor by inhibiting anodic and cathodic reactions. The adsorption of this inhibitor followed the Langmuir adsorption isotherm revealing high correlation coefficients and negative values of Gibbs free energy of adsorption showing that the inhibitor was adsorbed spontaneously as a monolayer on the surface of mild steel. Studies of the temperature-dependent polarization and thermodynamic analysis show a decrease in activation energy and increase in inhibition efficiency with temperature indicating a considerable chemisorption character of the adsorbed species. Differential capacitance and potential of zero charge measurements further confirmed the adsorption of inhibitor molecules by the formation of a stable protective film at the metal-solution interface. The combined experimental data prove that N-(pyridin-2-ylmethylene) aniline can prevent copper corrosion in acidic media. In protecting copper against sulphuric acid, the protection efficiency of N-(pyridin-2-ylmethylene) aniline is exceptionally high. The presence of nitrogen-containing adsorption centers and the conjugated π -electron system will enhance the interaction of the inhibitor on the copper surface.

Keywords: Copper corrosion; Schiff base; Corrosion inhibitor; Electrochemical impedance spectroscopy; Potentiodynamic polarization; Langmuir adsorption isotherm; Chemisorption.

1. INTRODUCTION

Copper is considered one of the most important engineering metals due to its excellent electrical conductivity, thermal conductivity, mechanical strength and atmospheric corrosion resistance. As a result of these qualities, it has become widely used in heat exchangers, condensers, cooling systems, electrical equipment, printed circuit boards and chemical processing industries. Although highly corrosion-resistant, copper quickly deteriorates in strongly acidic media, notably nitric acid and sulphuric acid, which are used in industrial applications like pickling, acid cleaning, metal finishing and related chemical manufacturing. The reactivity of

these acids increases the dissolution of copper and, therefore, damages and economic losses as well as decreasing the lifetime of industrial units (Revie and Uhlig et al. 2011; Fateh et al. 2020; Schweitzer et al. 2010).

Copper corrosion in nitric acid is predominately controlled by anodic dissolution of metal combined with cathodic reduction of nitrate ions whereas copper corrosion in sulphuric acid involves copper oxidation with hydrogen evolution and oxygen reduction reactions. Several factors play a crucial role in the rate of corrosion and these include concentration of acid, temperature level, dissolved generation of oxygen and adsorption of

aggressive ions at the metal/solution interface. Consequently, effective strategies for controlling corrosion remain an important area for materials engineers and industrial chemists research (Fateh *et al.* 2020; Antonijević and Petrović *et al.* 2008).

The application of organic corrosion inhibitors is one of the most economical and effective corrosion protection methods. Organic substances that contain nitrogen, oxygen, sulphur, and conjugated π -electron systems get adsorbed on metallic surfaces. This forms a dense protective film that inhibits anodic and cathodic electrochemical reactions. Generally, the inhibition efficiency depends on the molecular structure, electron density, adsorptivity and nature of interaction between the inhibitor molecules and the metal surface.

Schiff bases are a significant type of organic compound which is considered to have good capacity as corrosion inhibitor due to the presence of azomethine ($-C=N-$) functional group together with aromatic rings and maybe heteroatoms which can coordinate with vacant d-orbitals of transition metals. The structural features promote the absorption due to donor-acceptor interactions leading to the formation of stable protective films over metals. Many research works have shown that Schiff bases display excellent inhibition performances for steels, aluminium, copper and their alloys in acidic environments (Verma *et al.* 2018; Gece *et al.* 2008; Khaled *et al.* 2010; Xia *et al.* 2008).

Recent advances in the field of corrosion science have attracted the attention of researchers to the probable significance of synergetic experimental electrochemical techniques with molecular modelling techniques to understand inhibition mechanisms. Weight-loss measurements provide information about the corrosion rate and inhibition efficiency, while electrochemical impedance spectroscopy (EIS) and potentiodynamic polarization techniques reveal the charge transfer kinetics, behaviour of double layer and effect of inhibitors on anodic and cathodic reactions. Moreover, knowledge of adsorption isotherms and thermodynamic parameters gives information on spontaneity and nature of adsorption of inhibitor on metal surface (Bard and Faulkner *et al.* 2001; Mansfeld *et al.* 1990; McCafferty *et al.* 2010; Kokalj *et al.* 2010).

The HOMO and LUMO energies, energy gap, dipole moment, electronegativity, global hardness, softness and electrophilicity are some of the quantum chemical descriptors that help predict the adsorption tendency as well as electron transfer between the inhibitor molecules and the metal surfaces. The experimental results are being integrated with theoretical calculations, which would help in

understanding the corrosion inhibition mechanism for better design of organic inhibitors (Parr and Yang *et al.* 1989; Pearson *et al.* 1986; Obot *et al.* 2015; Khaled *et al.* 2022).

N-(2-pyridylmethylene) aniline was identified as a nitrogenous Schiff base containing an azomethine ($-C=N-$) link, a pyridine ring and an aromatic phenyl group which all possess favorable adsorption sites to interact with copper. The nitrogen lone-pair electrons and delocalized π -electron cloud would promote strong adsorption on the copper surface, thereby hindering the metal dissolution in aggressive acidic media. Paraphrasing this sentence will make the ideas of the original text clearer. Thus, various Schiff bases have been studied as corrosion inhibitors. However, little is known about N-(pyridin-2-ylmethylene) aniline and its corrosion inhibition behavior for copper in 1.0 M HNO_3 and 0.5 M H_2SO_4 .

The current study probes the capacity of N-(pyridin-2-ylmethylene) aniline to mitigate corrosion of copper immersed in nitric and sulphuric acid solutions. The weight-loss measurements, electrochemical impedance spectroscopy as well as potentiodynamic polarization are employed for this study. Various adsorption studies and thermodynamic evaluations are also done in this regard. The study aims to clarify the adsorption mechanism as well as to prove the effectiveness of this Schiff base (SB) as an efficient inhibitor for copper in an aggressive acidic medium, thus paving the way for high-performance organic inhibitors for industrial use.

2. MATERIALS AND METHODS

2.1 Composition of Test Material

The composition of the copper sample applied in the current study is as shown below. Cu-99.5%, Ni-0.003%, Al-0.019%, Mn-0.002% and Si-0.116%.

2.2 Sample Preparation

Weight loss measurements were done with copper coupons of dimensions $2.5\text{ cm} \times 1\text{ cm} \times 0.1\text{ cm}$. These samples were then polished using emery papers of grades 1/0, 2/0, 3/0, 4/0 and then lubricated with acetone. Both polarity and impedance studies were done with a Teflon coated cylindrical copper electrode of known size (0.2826 cm^2) with the compound, with both polarities present.

2.3 Preparation of Stock Solutions

2.3.1 Preparation of acid solutions

Sulphuric acid (1.25 M) and nitric acid (2.5 M) were prepared by diluting the analytical grade

concentrated acid with double distilled water in 1000 mL standard measuring flask and their concentrations were found through titration with the help of standard sodium carbonate solution.

2.3.2 Preparation of inhibitor solutions

Inhibitor stock solutions (5 mM) were prepared in the respective acid solutions in 100 mL standard measuring flasks. Test solutions with a given concentration of the inhibitor in acids were prepared by properly diluting the stock solution of the inhibitor.

2.3.3 Preparation of KCl, KBr, KI solutions

The samples of KCl, KBr, and KI (1.0 mM) were prepared using a 100 mL standard measuring flask in double distilled water.

2.4 Corrosion Monitoring Techniques

2.4.1 Weight Loss Measurements

The loss of weight was measured in line with the standard practice of American Society of testing and Material (ASTM). The copper coupons were placed to 2 h into 100 mL of the corrosive media with and without the inhibitor at room temperature (301±1K). The difference in weight of the sample prior to and after immersion was measured by ACCULAB electronic top loading balance with 0.1 mg readability/sensitivity. inhibition efficiency (Thanapackiam and Subramaniam et al. 2019).

$$IE = \frac{W - W'}{W} \times 100 \quad (1)$$

unrestricted uninhibited blocked inhibited.

2.4.2 Electrochemical impedance measurements

A confirmation period of 45 minutes was then allowed before an impedance measurement was done after the sample had been immersed in corrosive media and this was found to be enough to confirm that $E_{OCP}/VS(CCE)$ was adequate to attain a constant value. Potentiostat was used to carry out electrochemical impedance measurements. Frequency of AC was drifted (100 kHz to 10 mHz) and a single 10 mV peak to peak sine wave made up the exciting signal. The calculation of ~~preventive efficiencies~~ inhibition efficiency (IE) was based on expression (Thanapackiam and Subramaniam et al. 2019).

$$IE\% = \frac{R_{ct} - R_{ct}'}{R_{ct}} \times 100 \quad (2)$$

Where R_{ct} and R_{ct}' are the values of charge transfer resistance in ~~unblocked~~ uninhibited and ~~blocked~~ inhibited solutions, respectively.

2.4.3 Potentiodynamic Polarization measurements

Different electrochemical techniques can be used to make the evaluation of the corrosion prevention effect, and a more powerful method is potentiodynamic polarization. Based on this measurement, current density *versus* voltage is obtained which will give data concerning the electrochemical behaviour and the corrosion ~~demeanour~~ behaviour of copper in the presence of inhibitors. Potentiodynamic polarization curves were recorded at a potential sweep rate of 1.6 mVs⁻¹ through the identical cell arrangement. The E vs. $\log I$ polarization curves were obtained by scanning the potential from the cathodic region below E_{OCP} to ~~higher positive energy~~ the anodic region above E_{OCP} to determine the ~~voltages of corrosion~~ corrosion potential E_{corr} and Tafel slopes β_a and β_c . Cathodic and anodic polarity curves were plotted and tangents were drawn. The point of intersection was used to calculate two tangents i_{corr} and E_{corr} . Inhibitor efficiency is given as (Thanapackiam and Subramaniam et al. 2019).

$$IE\% = \frac{i_{corr} - i_{corr}'}{i_{corr}} \times 100 \quad (3)$$

2.5 Study of the Synergetic Effect

Synergism may be described as an important amplification of the inhibitory effect of a single inhibitor, although the efficacy of a second inhibitor is less when applied alone, through the addition of a small portion of another inhibitor.

The ~~meshing~~ synergistic effect of halides and organic compound inhibitors are widely covered in the literature (Hudson and Warning 1970). Homogeneity is discussed as the simultaneous adsorption of both the halides anion and organic cation on the metal surface (Thanapackiam and Subramaniam et al. 2019).

Impedance measurements were used to determine the ~~resonant-inhibitory~~ corrosion inhibition performance of the halide ions (Cl⁻, Br⁻, I⁻) and inhibitors used in copper corrosion in two acidic media. The experiment was conducted using 0.5 mM of KCl, KBr and KI solutions. In order to evaluate the synergism parameter, Aramkey and Hackerman (1969) equation was used (Thanapackiam and Subramaniam et al. 2019).

$$\begin{aligned} S_0 &= 1 - \theta_{1+2} - \theta_{1+2}' \\ S_1 &= 1 - I_{1+2} - I_{1+2}' \end{aligned} \quad (4)$$

by cation; θ'_{1+2} = surface ~~envelope~~ coverage.

2.6 Measuring the Potential of Zero Charge (pzc)

The electrostatic attraction between the ions of the inhibitor material or the dipole and the charged metal surface causes the physical adsorption. The charge at the

metal surface is affected by the free corrosion potential (E_{corr}) and the potential relative to zero charge (PZC).

The electrochemical impedance spectra were also obtained at different applied DC voltages at an AC frequency of 20 kHz. The open circuit copper potential determines the surface charge as given by the following equation (Thanapackiam and Subramaniam *et al.* 2019).

$$E_r = E_{\text{O.C.P.}} - E_{\text{PZC}} \quad (5)$$

E_r is the rational corrosion potential of the and E_{ocp} is the open circuit potential, E_{pzc} is the potential of zero charge.

2.7 Adsorption, Isotherms and Thermodynamic Parameters

Adsorption isotherm is a mathematical equation which is used to relate surface concentration of the adsorbed species to the mass concentration of the same at a constant temperature. Isotherm in adsorption provides the dependence of the surface coverage (i.e. amount of adsorption) with adsorption species and concentrations of the in the solution. ΔG , ΔH and ΔS could also be available using the proper diagram to identify the values of the adsorption of the inhibitor.

2.8 Effect of Temperature

To measure the potential of activation and to know the mechanism of absorption of the inhibitor on the copper surface, potentiodynamic polarization experiments were conducted in various temperatures of 35 to 55°C in the presence or absence of the inhibitor in the acidic solutions. Arrhenius equation can be used to calculate the rate of erosion depending on the temperature.

$$i_{\text{corr}} = A e^{-E_a/RT} \quad (6)$$

where i_{corr} is the corrosion current density, A is the Arrhenius pre-layer constant, R is the universal gas constant, E is the activation energy and T is the absolute temperature.

2.9 Synthesis of Schiff Base N-(pyridin-2-ylmethylene)aniline

The Schiff base N-(pyridin-2-ylmethylene)aniline was synthesized from the condensation reaction of equimolar amounts (0.01 mol) of pyridine-2-carboxaldehyde and aniline using ethanol as solvent. The reactants were mixed in ethanol and a small quantity of glacial acetic acid was added to the mixture as the catalyst. The mixture was refluxed for about 3–4 hours to produce maximum amount of

Schiff base. The resulting mixture was cooled to ambient temperature to obtain the crystallized product. The solid, which was filtered out, was collected and washed to remove impurities using cold ethanol. It was then purified by recrystallization. The compound was purified and obtained as yellow crystalline solid in good yield (Fig.1).

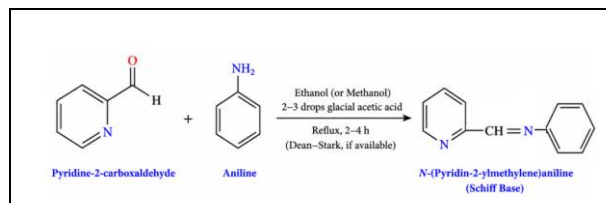


Fig.1 Synthesis of N-(Pyridin-2-ylmethylene)aniline

The formation of Schiff base was confirmed by the presence of vibrations representing $\text{C}=\text{N}$ at 1620 cm^{-1} . The peaks at 1590 cm^{-1} and 1515 cm^{-1} are related to the presence of the vibrations of $\text{C}=\text{C}$ and $\text{C}=\text{N}$ (aromatic rings) (Fig. 2).

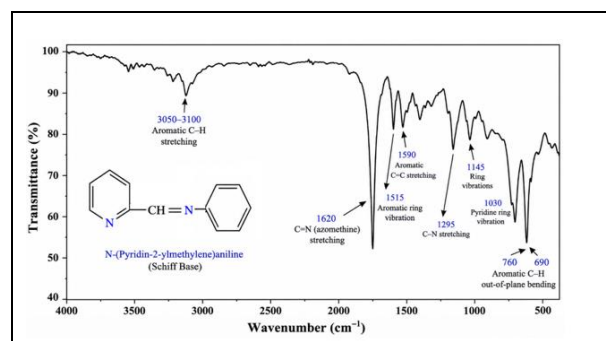


Fig.2 FTIR spectrum of Synthesized Schiff Base

3. RESULTS AND DISCUSSION

3.1 Weight Loss Measurement

Table 1 gives the performance on the corrosion inhibition of N-(pyridin-2-ylmethylene) aniline for copper in 1.0 M HNO_3 and 0.5 M H_2SO_4 solutions at various inhibitor concentrations. The corrosion rate of copper and inhibition efficiency increase with an increase in the concentration of the inhibitor. The results confirmed tests conducted in both acidic environments. This behavior suggests good adherence of inhibitor molecules onto the copper surface, thereby forming a barrier that retards the metal dissolution and the corrosion reaction.

The uninhibited solution contained 1.0 M HNO_3 in water showed a copper corrosion rate of 48.02 mmpy. The addition of the inhibitor at a concentration of 0.02

mM lowers the corrosion rate to 29.7 mmpy, which represents an inhibition efficiency of 36.0%. When the concentration of inhibitor was raised further, the corrosion rate further decreased and the corrosion rates were 22.23, 17.19 and 13.49 mmpy at 0.05, 0.10 and 0.50 mM, respectively. At a concentration of 1.00 mM, the corrosion rate was lowered to 10.80 mmpy with 77.5% efficiency. As the concentration increases, a greater number of inhibitor molecules were adsorbed onto the copper surface, which improved surface coverage and offered greater protection in the aggressive medium (nitric acid).

A comparable trend was noted in the 0.5 M H_2SO_4 medium. The corrosion rate of the blank solution was measured as 23.14 mmpy which is much lower than that observed in nitric acid indicating a comparatively high aggressiveness of HNO_3 towards copper. By adding an inhibitor at a concentration of 0.02 mM, the corrosion rate was reduced to 10.78 mmpy with an inhibition efficiency of 53.4%. Increasing the concentration of inhibitor to 0.05, 0.10 and 0.50 mM, the corrosion rate decreases further to 8.03, 4.98 and 3.77 mmpy, respectively. The maximum corrosion protection

was achieved at 1.00 mM, where the corrosion rate decreased tremendously to 1.46 mmpy. The inhibition efficiency was 93.7%. The significantly higher efficiency of inhibition in sulphuric acid as compared to nitric acid shows stronger adsorption and more stable film formation in H_2SO_4 medium.

The superior inhibiting performance of N-(pyridin-2-ylmethylene)aniline appears to be the result of the presence of nitrogen and sulphur heteroatoms and π -electron system in its molecule. The active centers allow for the adsorption on the copper surface, through donor-acceptor interactions with vacant orbitals of copper atoms (et al. 2000; Bentiss et al. 2001). The layer of inhibitor that forms help block active corrosion sites, and limits metal dissolution (anodic) and reduction (cathodic) [24-26]. The enhancement of inhibition efficiency with concentration demonstrates the effect of concentration on an adsorption process which takes place with the formation of a compact protective film at high inhibitor concentrations (Silverstein et al. 2005).

Table.1 Corrosion inhibition efficiency of N-(pyridin-2-ylmethylene) aniline on corrosion of copper in higher concentration of 1.0 M HNO_3 and 0.5 M H_2SO_4 solution

Medium	Inhibitor concentration (mM)	Corrosion rate (mmpy)	Inhibitor efficiency (%)
1.0 M HNO_3	Blank	48.02	–
	0.02	30.35	36.8
	0.05	22.23	53.7
	0.10	17.19	64.2
	0.50	13.49	71.9
	1.00	10.80	77.5
0.5 M H_2SO_4	Blank	23.14	–
	0.0	10.78	53.4
	0.05	8.03	65.3
	0.10	4.98	78.5
	0.50	3.77	83.7
	1.00	1.46	93.7

A comparison of both acidic media indicates that the inhibitor shows better inhibition performance in 0.5 M H_2SO_4 as compared to 1.0 M HNO_3 . This difference is attributed to the strong oxidizing ability of nitric acid, which disrupts the adsorbed layer of inhibitor to some extent and increases copper dissolution. On the other hand, sulphuric acid can provide conditions that favor the adsorption of inhibitor molecules becoming stronger. As a result, higher inhibition efficiencies were achieved in H_2SO_4 medium.

The observations indicate that copper corrosion can be prevented *via* the use of N-(pyridin-2-ylmethylene) aniline. With the increase in concentration, the inhibition efficiency increases systematically. It reaches maximum values of 77.5% in nitric acid and 93.7% in sulphuric acid at 1.00 mM concentration. These results show the applicability of as a corrosion-control compound for copper in aggressive acidic media.

3.2 Electrochemical Impedance Spectroscopy

The corrosion behavior of copper in 1.0 M HNO_3 and 0.5 M H_2SO_4 solution with various concentrations of synthesized Schiff base inhibitor was investigated using Electrochemical Impedance Spectroscopy (EIS). Table 2 shows a summary of the values of impedance parameters obtained from the Nyquist plots, whilst the Nyquist plots are shown in Figs 6.3a and b. The spectra exhibit depressed capacitive semicircles accompanied by low-frequency Warburg diffusion tails, which imply that the corrosion process is controlled by charge-transfer and mass-transport processes (Pavia *et al.* 2009; Coates *et al.* 2006; Nakamoto *et al.* 2009).

Copper exhibited relatively small semicircle diameters indicating low charge-transfer resistance and rapid corrosion in the absence of inhibitor in the acidic media. The R_{ct} value of blank solution in 1.0 M HNO_3 is $89.4 \Omega \text{ cm}^2$ whereas R_{ct} value of that in 0.5 M H_2SO_4 is $65.8 \Omega \text{ cm}^2$. As the Schiff base is added, the diameter of the semicircle increased progressively, which indicates an increase in charge-transfer resistance and a decrease in corrosion rate. The adsorption of inhibitor molecules on copper surface suppresses the metal dissolution reactions from taking place, as it is evident from this behavior (El-Etre *et al.* 2003)

The charge-transfer resistance for nitric acid system gradually increases from $116.8 \Omega \text{ cm}^2$ at 0.02 mM concentration of inhibitor to $364.8 \Omega \text{ cm}^2$ at 1.00 mM. Inhibition effectiveness increased from 24% to 76% at the same time. The increase in R_{ct} indicates that the inhibitor successfully blocks the active corrosion sites and impedes the electron transfer between the metal-solution interface. The growing extent of the Nyquist semicircles illustrated in Fig. 3a provides additional

evidence indicating the presence of a protective adsorbed layer on copper.

The constant phase element admittance (Y_0) decreased from $845.6 \times 10^{-6} \Omega^{-1} \text{ cm}^{-2}$ for the uninhibited solution to $176.2 \times 10^{-6} \Omega^{-1} \text{ cm}^{-2}$ at 1.00 mM inhibitor concentration. The decrease in the value of constant phase element admittance (Y_0) indicates a reduction in the local dielectric constant and an increase in the thickness of the electrical double layer, resulting from the adsorption of inhibitor molecules onto the metal surface (El-Etre *et al.* 2003). Similarly, the double-layer capacitance (C_{dl}) decreased from 144.2 to $62.9 \mu\text{F cm}^{-2}$. Such a reduction is generally attributed to the displacement of water molecules by adsorbed inhibitor species and the formation of a compact protective film that effectively covers the metal surface, thereby enhancing corrosion resistance (El-Etre *et al.* 2003; Obot and Obi-Egbedi *et al.* 2010).

Table.2 Electrochemical impedance parameters of copper corrosion in 1.0 M HNO_3 and 0.5 M H_2SO_4 solutions in the absence and presence of N-(Pyridin-2-ylmethylene) aniline at various concentrations.

Medium	Inhibitor Concentration. (mM)	$R_{ct} (\Omega \text{ cm}^2)$	$Y_0 \times 10^{-6} \Omega^{-1} \text{ cm}^{-2}$	n	Cdl ($\mu\text{F cm}^{-2}$)	IE (%)
HNO_3	Blank	89.4	845.6	0.98	144.2	–
	0.02	116.8	670.4	0.95	121.0	24
	0.05	158.7	395.2	0.94	116.4	44
	0.10	194.6	188.5	0.93	92.8	55
	0.50	301.5	184.9	0.92	77.2	71
H_2SO_4	1.00	364.8	176.2	0.94	62.9	76
	Blank	65.8	325.7	0.93	156.5	–
	0.02	188.6	258.9	0.95	126.4	53
	0.05	251.4	231.8	0.95	109.5	65
	0.10	394.2	179.6	0.94	96.4	78
	0.50	835.7	149.8	0.93	80.3	90
	1.00	1088.0	141.7	0.93	69.8	92

Fig.3a Electrochemical impedance spectra of corrosion of copper in 1.0 M HNO₃ in the absence and presence of N-(pyridin-2-ylmethylene) aniline at varying concentrations

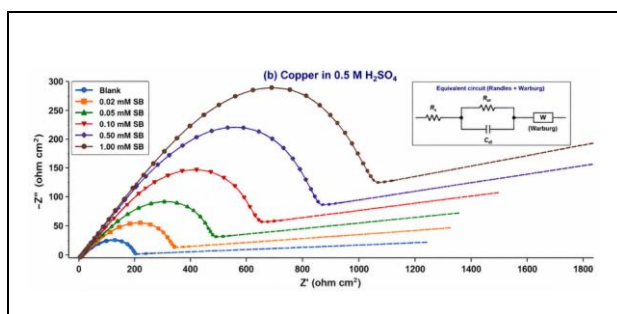


Fig.3b Electrochemical Impedance Spectra of corrosion of copper in 0.5M H₂SO₄ in the absence and presence of N-(Pyridin-2-ylmethylene)aniline at varying concentrations

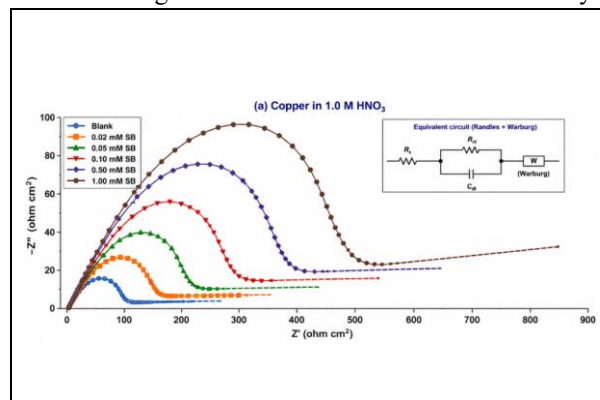
A comparable tendency was noted in sulphuric acid medium but inhibition performance was much higher. The charge transfer resistance increased drastically after the addition of an inhibitor, rising from 65.8 $\Omega \text{ cm}^2$ for the blank solution to 1088.0 $\Omega \text{ cm}^2$ at the 1.00 mM concentration. The inhibition efficiency increased from 53% to 92%. This shows that the Schiff base acts as an effective protective agent in sulphuric acid. As depicted in Fig. 6.3b, a significant expansion in the diameter of the Nyquist semicircle indicated strong adsorption of inhibitor molecules, which inhibited corrosion reactions.

The Y_0 values in sulphuric acid were observed to decrease from 325.7×10^{-6} to $141.7 \times 10^{-6} \Omega^{-1} \text{ cm}^{-2}$ as the concentration of inhibitor increased. C_{dl} also decreased from 156.5 to 69.8 $\mu\text{F cm}^{-2}$. The nature of interfacial region changes significantly due to the adsorption of inhibitor molecules. The effectiveness of the double layer was increased and the active surface area for corrosion was decreased due to the formation of an adsorbed film.

The values of the heterogeneity factor (n) were found close to unity in both the media, which were in the range 0.92 to 0.98. The copper surface did not seem to lose its capacitive character with the adsorption of inhibitor molecules. Deviations from ideal capacitor behaviour are likely due to the surface roughness, non-uniform adsorption and microscopic heterogeneities at the electrode surface (El-Etre *et al.* 2003; Obot and Obi-Egbedi *et al.* 2010).

A noteworthy characteristic of Fig. 3a and 3b is the Warburg impedance which begins to appear at low frequencies and is shown by the inclined diffusion tails which follow the capacitive semicircles. The Warburg

impedance suggests that the inhibitor being studied diffuses through both the solution and the adsorbed layer



attached to the metal surface (Ebenso *et al.* 2004; Lebrini *et al.* 2005). As the concentration of the inhibitor rises, a more effective restriction overall is observed in the diffusion paths since a denser protective film quickly forms. As a result, the mass transport of aggressive ions to the copper surface becomes further hindered, which leads to improved corrosion resistance.

On comparing the performances of Schiff base in two acidic media, the results show that in 0.5 M H₂SO₄ offers better inhibition than 1.0 M HNO₃. The maximum inhibition efficiency obtained in sulphuric acid (92%) is more than that of nitric acid (76%). This difference is due to the highly oxidizing nature of nitric acid capable of partially destabilizing the adsorbed inhibitor layer and enhancing copper dissolution. Unlike the previous case, favorable conditions are generated by sulphuric acid for stable adsorption and compact film formation (Obot and Obi-Egbedi *et al.* 2010).

The strong inhibition performance of Schiff base can be justified by the presence of nitrogen and sulphur donor atoms along with conjugated π -electron systems (Obot and Obi-Egbedi *et al.* 2010; Ebenso *et al.* 2008; Khaled *et al.* 2010). The inhibitor molecules adsorb on copper through interaction with vacant copper orbitals of these adsorption centers which are responsible for the development of a strongly adherent protective layer. The adsorbed film serves as a physical and electrochemical barrier, inhibiting anodic copper dissolution and cathodic reduction reactions (Verma *et al.* 2017; Popova *et al.* 2007).

To summarize, the results of EIS indicate that the introduction of Schiff base is an efficient copper corrosion inhibitor in both acidic media. The adsorption-controlled inhibition mechanism is confirmed from rise in charge-transfer resistance, drop in double-layer capacitance, reduction in constant phase element admittance values and Warburg diffusion behavior. At a concentration of 1.00 mM, the inhibitor gives maximum protection with better performance in sulphuric acid medium due to the formation of a more stable and compact protective film.

3.4 Potentiodynamic Polarization Studies

For studying the corrosion behavior of copper in 1.0 M HNO_3 and in 0.5 M H_2SO_4 , the potentiodynamic polarization technique was employed in the absence and presence of various concentrations of Schiff base inhibitor. The electrochemical parameters obtained from the polarization curves are shown in Table 3. The Tafel plots are displayed in Fig. 4a and 4b. The polarization data can be useful for determining values such as corrosion current density (I_{corr}), corrosion potential (E_{corr}), anodic and cathodic Tafel slopes (β_a and β_c), corrosion rate, and inhibition efficiency.

The I_{corr} of the blank solution was determined to be $111.2 \mu\text{A cm}^{-2}$ which corresponded to a corrosion rate of 179.2 mpy indicating severe corrosion of copper in nitric acid medium. In the presence of Schiff base inhibitor, considerable reduction in the corrosion current density occurred. The I_{corr} values of 0.0004 mM exhibited the value of $82.7 \mu\text{A cm}^{-2}$ with inhibition efficiency of 25%. Continued addition of inhibitor decreased the current density more and more and the current density was $24.6 \mu\text{A cm}^{-2}$ at 0.02 mM. At the same time, the corrosion rate dropped from 179.2 to 39.8 mpy while the inhibition efficiency increased to 78%.

The significant decrease of corrosion current density reveals that this Schiff base effectively inhibits the electrochemical dissolution of copper (Ebenso *et al.* 2010). As the concentration of the inhibitor increased, the corrosion rate decreased, signifying the increased surface coverage by the adsorbed inhibitor molecules. The molecules form a barrier between the metal surface and the aggressive acid that diminishes the rate of charge transfer and metal dissolution (Solmaz *et al.* 2010).

The E_{corr} was shifted from 40.85 mV for the uninhibited solution to -19.10 mV at maximum of inhibitor concentration. Although a negative shift in E_{corr} is noted, the displacement magnitude is less than 85 mV compared to blank solution (Obot and Obi-Egbedi *et al.* 2010). Based on the electrochemical criteria, it can be concluded that the Schiff base N-(pyridin-2-ylmethylene) aniline behaves as a mixed-type inhibitor because of the shift of the anodic and cathodic peaks in

opposite directions without preferential effect on either process.

Changes in anodic and cathodic Tafel slopes were noticed with the addition of inhibitor. The change in values of both β_a and β_c shows that the inhibitor affects the kinetics of both the anodic and cathodic reactions occurring on copper. This observation further supports the mixed-mode inhibition mechanism.

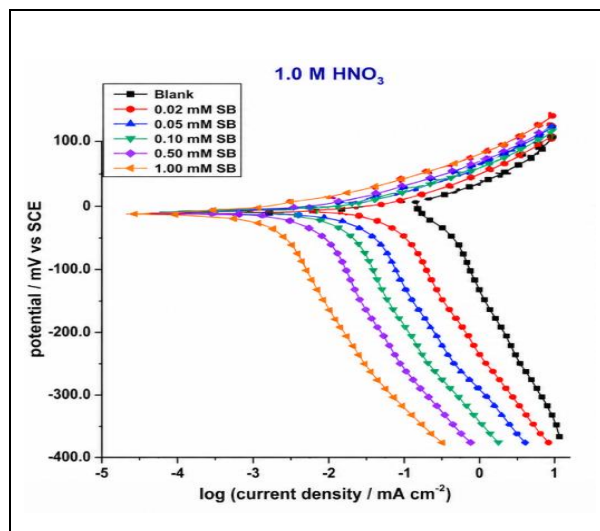


Fig.4a Tafel plots of copper corrosion in 1.0 M HNO_3 in presence and absence of N-(pyridin-2-ylmethylene) aniline at various concentrations

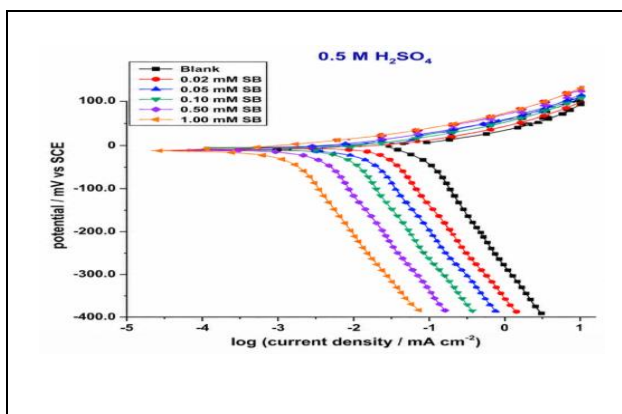


Fig.4b Tafel plots of copper corrosion in 0.5 M of H_2SO_4 in presence and absence of N-(pyridin-2-ylmethylene) aniline at various concentrations

Table.3: Corrosion of copper in 1.0 M HNO₃, and 0.5 M H₂SO₄ acid solutions in the absence and presence of N-(Pyridin-2-ylmethylene) aniline at various concentrations in electrochemical parameters.

Medium	Concentration (mM)	E _{corr} (mV)	I _{corr} (μA cm ⁻²)	β _c (mV dec ⁻¹)	β _a (mV dec ⁻¹)	Corrosion Rate (mpy)	IE (%)
1.0 M HNO₃	Blank	40.85	111.2	358.4	64.8	179.2	–
	0.0004	8.10	82.7	208.5	68.4	133.5	25
	0.001	-13.25	44.6	165.8	65.9	71.8	60
	0.002	-10.85	39.4	151.2	54.7	63.7	64
	0.01	-11.60	28.2	259.4	48.5	45.2	75
	0.02	-19.10	24.6	139.8	51.6	39.8	78
0.5 M H₂SO₄	Blank	5.20	24.1	308.6	50.2	38.9	–
	0.0004	-25.40	17.8	368.2	48.9	28.7	27
	0.001	-28.60	11.9	389.5	46.8	19.1	51
	0.002	-41.75	6.95	594.2	43.7	11.2	71
	0.01	-34.10	5.35	248.7	48.4	8.65	78
	0.02	-30.40	4.02	218.6	47.5	6.48	83

The polarization curves in sulphuric acid have shown similar trends. However, the inhibition was much greater. The I_{corr} value for blank solution was 24.1 μA cm⁻² having a corrosion rate of 38.9 mpy. The addition of the Schiff base decreased the corrosion current density gradually (17.8 μA cm⁻² at 0.0004 mM to 4.02 μA cm⁻² at 0.02 mM concentration).

The corrosion rate of copper decreased from 38.9 to 6.48 mpy and the inhibition efficiency increased from 27% to 83%. The significant reduction of the corrosion current density shows the good protective performance of the Schiff base in sulphuric acid medium. The presence of much lower current density values at a high concentration of inhibitor is due to the formation of a compact and highly adherent film.

The corrosion potential in blank solution was found to be +5.20 mV and in higher inhibited concentration, it was -30.40 mV. Like the nitric acid system, the shift in E_{corr} remained below the criterion for being classified as a purely anodic or cathodic inhibitor. As a result, the Schiff base can also be classified as a mixed-type inhibitor in sulphuric acid medium.

Fig. 4b illustrates that with the increase in concentration of the inhibitor, the cathodic and anodic branches of Tafel plot move towards lower current density values. This behaviour demonstrates that adsorption of inhibitor molecules on the metal surface effectively inhibited both anodic copper dissolution and cathodic hydrogen evolution reactions (Obot and Obi-Egbedi et al. 2010).

The polarization curves for copper in nitric acid solution (Fig.4a) and sulphuric acid solution are shown in Fig. 4b. The polarization curves were systematically shifted toward lower current density regions with increasing inhibitor concentration in both the media. As the concentration of the inhibitor increased, the distance between the anodic and cathodic branches increased progressively. The slope decrease of the anodic branch implies that copper oxidation kinetics slowed down while the changes on the cathodic branch imply suppression in reduction reactions on the metal surface. According to some polarization curves being approximately parallel, the fundamental mechanism of corrosion is not altered (Macdonald *et al.* 2005). The inhibitor, in essence, prevents active sites from performing and reduces the effective area of the working electrode.

The presence of nitrogen and sulphur atoms along with conjugated π -electron systems in the molecular structure of the Schiff base is responsible for its inhibition behavior (Obot and Obi-Egbedi *et al.* 2010). Through donor-acceptor interactions with vacant copper orbitals, copper surface adsorption occurs at the electron-rich centers (Mansfeld *et al.* 1990). As the concentration of the inhibitor rises, a greater number of the molecules will get adsorbed, enhancing surface coverage and a compact protective film will form.

The adsorbed layer restricts the diffusion of aggressive ions as well as inhibits electron transfer reactions at the metal-electrolyte interface. As a result, corrosion current densities, corrosion rates and inhibition efficiencies are lower. The effectiveness of the inhibitor in both acidic environments manifests the role of adsorption as the principal corrosion protection mechanism.

As evidenced from the comparative inhibition efficiency, the Schiff base was found to be more effective in 0.5 M H_2SO_4 medium where a maximum of 83 % inhibition efficiency was achieved compared with 78 % in 1.0 M HNO_3 . The stronger adsorption and higher stability of the inhibitor film in sulphuric acid enhances its protection efficiency. The inhibition efficiency is lower because the highly oxidizing nature of nitric acid may partially destabilize the adsorbed layer (Barsoukov and Macdonald *et al.* 2005; Khaled *et al.* 2008).

6.5 Adsorption Isotherm

The adsorption isotherm analysis was used to study the adsorption behaviour of N-(pyridin-2-ylmethylene) aniline on copper in 1.0 M HNO_3 and 0.5 M H_2SO_4 . The relationship between C and θ was studied in order to ascertain the mode of adsorption of the inhibitor as well as the interaction of the inhibitor molecules on the copper surface. Table 6.4 presents the adsorption parameters derived from the Langmuir

adsorption model and the adsorption isotherms from the same model are shown in Fig. 5a and 5b.

Fig. 5a shows the adsorption isotherm for copper in a 1.0 M HNO_3 solution. The plot shows excellent linearity with the regression equation of $y = 0.9997x + 0.0974$.

$$C/\theta = 1.2654C + 0.0571; R^2 = 0.9997.$$

The high value of R^2 suggests that the adsorption is well-described by the Langmuir adsorption model with monolayer adsorption of inhibitor on the copper surface (Popova and Christov *et al.* 2006; Bentiss *et al.* 2002). The slope value is quite close to 1 indicating that there is negligible interaction between the adsorbed species and that the adsorption sites are uniformly distributed on the surface of the metal (Obot and Ebenso *et al.* 2010; Noor *et al.* 2009; Finsgar and Jackson *et al.* 2014; Khamis *et al.* 1990).

The calculated adsorption equilibrium constant ($K_{\text{ads}} = 17.50$) is indicative of the appreciable affinity of inhibitor molecules for copper. The standard free energy of adsorption value was negative ($\Delta G_{\text{ads}}^\circ = -17.33 \text{ kJ mol}^{-1}$), which indicated that the adsorption process was spontaneous. When the value of $\Delta G_{\text{ads}}^\circ$ is less negative than -20 kJ mol^{-1} , then it suggests that the adsorption occurs mostly through physical adsorption, and for example, there is electrostatic attraction between the charged inhibitor molecules and the metal surface.

In nitric acid medium (1.0 M), N-(pyridin-2-ylmethylene) aniline is adsorbed on copper surface forming an inhibitive molecular layer which shields active sites of corrosion and minimizes direct contact between copper and aggressive nitrate ions. Therefore, with increase in concentration of inhibitor, the rate of corrosion decreases, while inhibition efficiency increases.

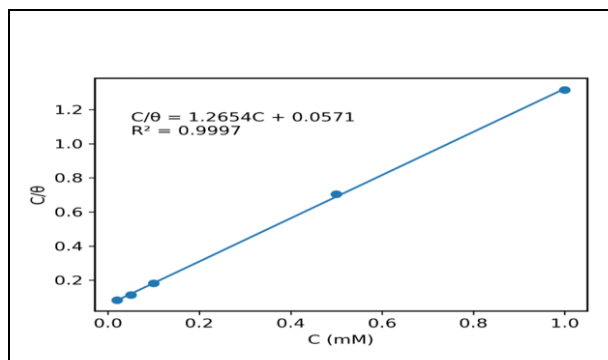


Fig.5a Isotherm of adsorption of N-(pyridin-2-ylmethylene) aniline on copper in 1.0 M HNO_3 acid solution

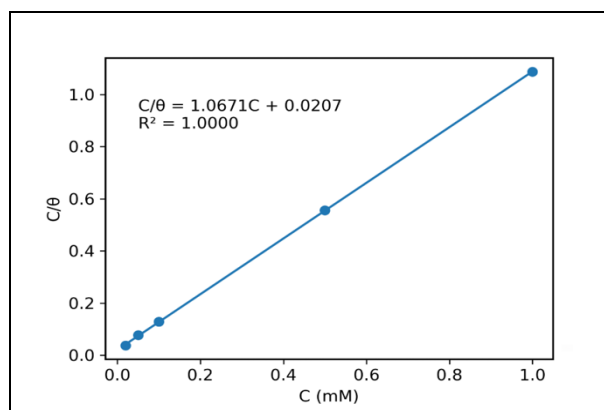


Fig.5b Isotherm of adsorption of N-(pyridin-2-ylmethylene) aniline on copper in 0.5 M H₂SO₄ acid solution

Table.4 Adsorption isotherm parameters for corrosion of copper in 1.0 M HNO₃, and 0.5 M H₂SO₄ acid solutions in the absence and presence of N-(pyridin-2-ylmethylene)aniline at various concentrations

Medium	Linear Equation (C/θ vs C)	R ²	K _{ads}	ΔG° _{ads} (kJ mol ⁻¹)
C/θ =				
HNO ₃	1.2654C + 0.0571	0.9997	17.50	-17.33
C/θ =				
H ₂ SO ₄	1.0671C + 0.0207	0.99996	48.40	-19.89

The adsorption isotherm of copper in 0.5 M H₂SO₄ is shown in Fig.5b. The obtained linear relationship is given by

$$C/\theta = 1.0671C + 0.0207.$$

with an incredibly high correlation coefficient ($R^2 = 1.0000$).

The straight line indicates good conformity with the Langmuir model. In addition, the slope value being close to unity indicates inhibitor molecules are adsorbed in monolayer form on homogeneous copper surfaces (Stern and Geary *et al.* 1957; Ferreira *et al.* 2004). The strong conformity to the Langmuir model suggests that only one inhibitor can accumulate at one site without lateral interaction between adsorbed species (Ebenso *et al.* 2003).

The value of the adsorption equilibrium constant (K_{ads}) observed was 48.40 which is higher than that obtained in nitric acid medium. It means that the higher value indicates stronger adsorption of inhibitor molecules and great affinity towards the copper surface in sulphuric acid solution. The stronger adsorption contributes to the superior corrosion inhibition in H₂SO₄ medium.

The standard free energy of adsorption value of -19.89 kJ mol⁻¹ is more negative than that corresponding to nitric acid. The negative value indicates the spontaneity of the adsorption process. The magnitude of ΔG°_{ads} indicates that adsorption mainly involves physisorption, although some strong interaction between the inhibitor molecules and the copper substrate has occurred. This higher negative free energy value indicates that a more stable adsorbed film formed in sulphuric acid solution.

Comparison of the adsorption parameters revealed significant differences between the two acidic media. The adsorption equilibrium constant (K_{ads}) increased from 17.50 in HNO₃ medium to 48.40 in H₂SO₄ medium, indicating stronger adsorption of the inhibitor molecules on the copper surface in sulphuric acid. Furthermore, the standard free energy of adsorption (ΔG°_{ads}) became more negative, changing from -17.33 kJ mol⁻¹ in nitric acid to -19.89 kJ mol⁻¹ in sulphuric acid. These results suggest that the adsorption process is more favorable and spontaneous in sulphuric acid medium, leading to the formation of a more stable and protective adsorbed inhibitor layer (El-Etre *et al.* 2003).

The observations indicated that the inhibitor molecules form a stable and compact protective coating on copper in sulphuric acid compared to nitric acid. Nitric acid may be partially acting as a hindrance to adsorption and film stability because of its very oxidizing behavior. Sulphuric acid may provide a better environment for the adsorption of the inhibitor (Fouda and El-Haddad *et al.* 2009). Inhibition efficiencies in most cases are found to be higher in H₂SO₄ medium.

The presence of electron-rich nitrogen atoms and conjugated π -electron systems of Schiff base are responsible for the adsorption process. These adsorption sites interact with the copper interface through electrostatic attraction, donor-acceptor interaction and partial electron sharing (Noor and Al-Moubaraki *et al.* 2008). After they are adsorbed, the inhibitor molecules replace the water molecules that are initially adsorbed on the metal surface. Thus, they create a compact protective layer, blocking the sites of active corrosion (Oguzie *et al.* 2008).

Evidence of spontaneity can be seen from negative values of ΔG°_{ads} which confirm the thermodynamic feasibility of the adsorption process (El-

Etre *et al.* 2003; Obot and Obi-Egbedi *et al.* 2010). A good fit to the Langmuir model indicates that corrosion protection occurs through the formation of a uniform monolayer of inhibitor molecules on copper surfaces (Ebenso and Obot *et al.* 2010).

3.6 Effect of Temperature

The potentiodynamic polarization measurements were used to study the effect of temperature on the corrosion behaviour of copper in 1.0 M HNO₃ and 0.5 M H₂SO₄, without and with 0.02 mM Schiff base inhibitor. Table 5 presents the electrochemical parameters derived from the Tafel plots, while the polarization curves are illustrated in Fig. 6.6a–6.6d. The investigation was carried out at temperatures 308, 318 and 328 K to study the thermal stability and inhibition ability of the Schiff base at various operating conditions.

As demonstrated in Fig. 6a, the corrosion of copper in nitric acid medium is faster with increasing temperature. The corrosion current density (I_{corr}) was 31.2 $\mu\text{A cm}^{-2}$ at 308 K, 203.5 $\mu\text{A cm}^{-2}$ at 318 K, and 364.7 $\mu\text{A cm}^{-2}$ at 328 K. You can refer to stain (metal) according to the material of the given metal (the stain may be female or male or other).

As the temperature increases, the corrosion current density (I_{corr}) also increases, indicating enhanced electrochemical activity at the copper–solution interface (Verma *et al.*, 2017). The rise in temperature increases the kinetic energy of the reacting species, thereby facilitating the dissolution of copper and resulting in a higher corrosion rate. Furthermore, increasing temperature causes both the anodic and cathodic branches of the polarization curves to shift toward higher current densities, demonstrating that both the metal oxidation and cathodic reduction reactions are accelerated (El-Etre *et al.* 2003). These observations confirm that elevated temperatures promote the overall corrosion process by enhancing the kinetics of the electrochemical reactions occurring on the copper surface.

The electrochemical equilibrium associated with the corrosion process is affected by temperature as shown by the slight increase in the corrosion potential (E_{corr}) value from 3.10 mV at 308 K to 22.8 mV at 328 K. Nonetheless, the small changes of E_{corr} indicate the change of corrosion mechanism is not altered essentially.

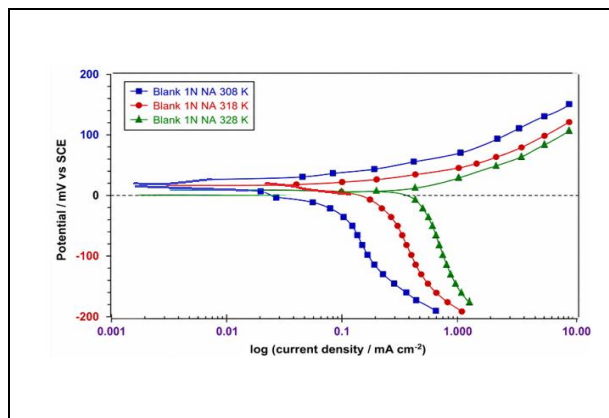


Fig.6a Potentiodynamic polarization behavior of copper in 1.0 M nitric acid solution at 308, 318, and 328 K in the absence of N-(pyridin-2-ylmethylene) aniline

Fig. 6b shows the polarization behavior of copper in nitric acid containing 0.02 mM Schiff base inhibitor. The corrosion current density observed with the inhibitor was always much lower than that of the uninhibited system at all temperatures. The I_{corr} values showed an increase from 15.4 $\mu\text{A cm}^{-2}$ at 308 K to 74.2 $\mu\text{A cm}^{-2}$ at 318 K and to 97.6 $\mu\text{A cm}^{-2}$ at 328 K.

Corrosion continued to rise with temperature, but the rate of increase was much less than in blank solution. The efficiency of inhibition increased from 51% at 308 K to 64% at 318 K and 73% at 328 K. An increase in inhibition efficiency with increasing temperature indicates that adsorption of the inhibitor becomes more favorable.

Enhancement of inhibition performance at higher temperatures could be attributed to stronger interaction between inhibitor molecules and the surface of copper (Solmaz and Kardas *et al.* 2008). At higher temperature, adsorption can be enhanced and a more compact protective film can be formed that constrains charge transfer and copper dissolution (Yurt and Bereket *et al.* 2005). The Tafel curves indicate that both the anodic and cathodic branches shift to lower values of current density in the presence of an inhibitor. This behavior shows that Schiff base is a mixed-type inhibitor (El-Etre *et al.* 2003; Obot and Obi-Egbedi *et al.* 2010). It inhibits anodic metal dissolution as well as cathodic reduction reaction.

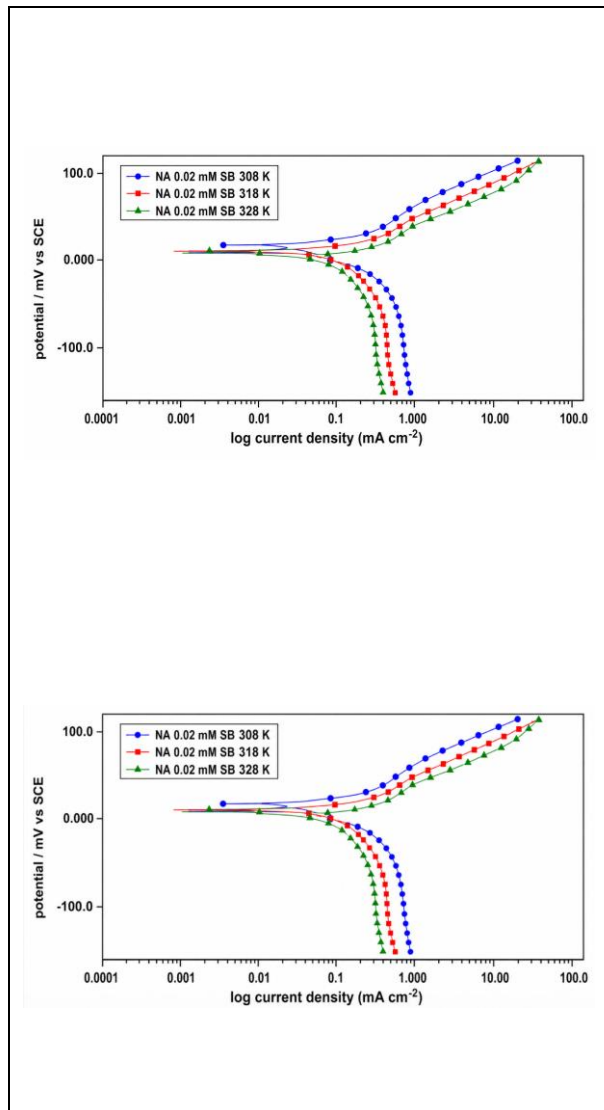


Fig.6b: Potentiodynamic polarization behavior of copper in 1.0 M nitric acid solution at 308, 318, and 328 K in the presence of N-(pyridin-2-ylmethylene) aniline

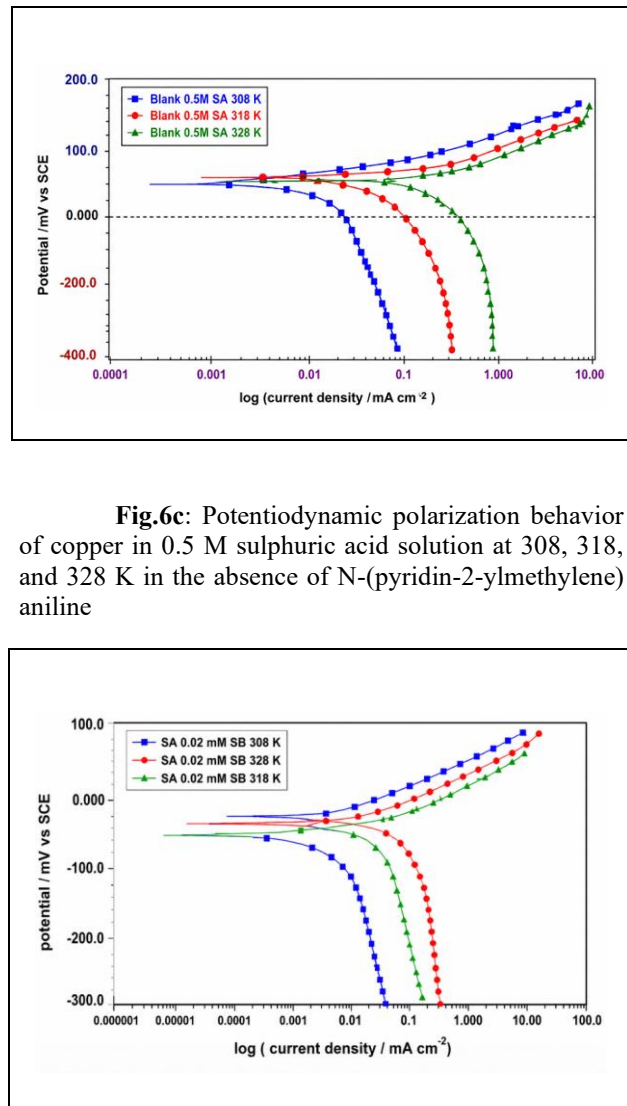


Fig.6c: Potentiodynamic polarization behavior of copper in 0.5 M sulphuric acid solution at 308, 318, and 328 K in the absence of N-(pyridin-2-ylmethylene) aniline

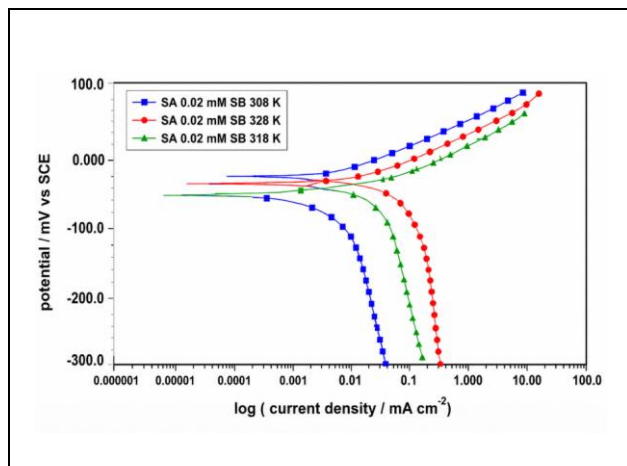


Fig.6d: Potentiodynamic polarization behavior of copper in 0.5 M sulphuric acid solution at 308, 318, and 328 K in the presence of N-(Pyridin-2-ylmethylene) aniline.

Table.4: Potentiodynamic polarization parameters of copper in 1.0 M HNO₃ and 0.5 M H₂SO₄ solutions with and without N-(pyridin-2-ylmethylene) aniline at various temperatures

Medium	Concentration (mM)	Temperature (K)	E _{corr} (mV)	I _{corr} (μA cm ⁻²)	β _c (mV dec ⁻¹)	β _a (mV dec ⁻¹)	Corrosion Rate (mpy)	IE (%)
HNO ₃	Blank	308	3.10	31.2	176.8	55.2	50.5	–
	Blank	318	11.4	203.5	418.2	66.9	329.8	–
	Blank	328	22.8	364.7	446.5	60.8	590.1	–
	0.02 mM	308	7.05	15.4	368.4	78.5	24.9	51

Medium	Concentration (mM)	Temperature (K)	E_{corr} (mV)	I_{corr} ($\mu\text{A cm}^{-2}$)	β_c (mV dec^{-1})	β_a (mV dec^{-1})	Corrosion Rate (mpy)	IE (%)
H₂SO₄	0.02 mM	318	9.35	74.2	337.6	62.4	120.2	64
	0.02 mM	328	15.2	97.6	421.8	64.1	158.0	73
	Blank	308	-13.4	12.8	544.2	50.0	20.8	–
	Blank	318	-45.2	31.8	286.7	51.1	51.5	–
	Blank	328	12.6	91.4	278.5	63.4	147.8	–
	0.02 mM	308	-53.6	7.35	108.5	46.1	15.1	43
	0.02 mM	318	-30.7	13.1	224.6	47.2	21.2	59
	0.02 mM	328	-38.4	17.0	198.4	47.1	27.5	81

As shown in Fig. 6c, the polarization curves indicate that the corrosion rate of copper in sulphuric acid medium increases with temperature. The current density of corrosion increases from a value of 12.8 $\mu\text{A cm}^{-2}$ at 308K temperature to 31.8 $\mu\text{A cm}^{-2}$ at 318 K and 91.4 $\mu\text{A cm}^{-2}$ at 328 K. The corrosion rate corresponding to the behavior was found to be 20.8 mpy, 51.5 mpy and 147.8 mpy. Corrosion of copper increases with increasing temperature. Increased reaction kinetics, as well as enhanced mobility of aggressive ionic species in the electrolyte, cause the corrosion current density to increase. As we go on increasing the temperature, we find that the anodic and cathodic branches become increasingly steeper. This is suggestive of an acceleration of various electrochemical reactions at the important metal surface. Thus, a rise in temperature causes more copper to be dissolved and more cathodic processes occur, which overall leads to greater corrosion.

Fig.6d shows the effect of temperature on the inhibited sulphuric acid system. The corrosion current density was lowered at all temperatures by Schiff base. The I_{corr} values obtained at 308, 318 and 328 K were 7.35, 13.1 and 17.0 $\mu\text{A cm}^{-2}$, respectively.

The corrosion rate underwent a moderate increase, rising from 15.1 mpy at 308 K to 21.2 mpy at 318 K and to 27.5 mpy at 328 K. Nonetheless, the values were still below those in uninhibited medium by large margins. The inhibition performance increased from 43% at 308 K to 59% at 318 K, then reached the peak of 81% at 328 K.

The increase in inhibition efficiency with temperature indicates stronger chemisorptive interaction of Schiff base molecules with the copper surface. At higher temperature, adsorption becomes stronger and

stable, leading to the formation of a strong protective film against corrosion.

The polarization curves show lower current densities in the whole potential range indicating that the inhibitor suppresses anodic and cathodic reaction effectively. The fluctuations seen in both branches further validate the identification of the Schiff base as a mixed-type inhibitor.

The increased inhibition efficiency observed with increasing temperature in both acid media provides an indication of the adsorption mechanism. In many corrosion systems, the increase in temperature leads to desorption of physically adsorbed inhibitor molecules and causes a decrease in inhibition efficiency. In the current study, however, the reverse trend is found.

The increased corrosion inhibition efficiency from 51 in HNO_3 and 43 in H_2SO_4 to 73 in HNO_3 and 81 in H_2SO_4 shows that adsorption becomes stronger with increasing temperature. The inhibitor molecules form coordinate bonds with the metal surface by donating electron pairs from nitrogen atoms and the π -electron system of the Schiff base highlight a strong chemisorption (Obot and Obi-Egbedi *et al.* 2010). This increase in the surface coverage, compactness of the film, and blocking of active corrosion sites at elevated temperatures promotes higher adsorption. Consequently, the inhibitor remains effective under severe corrosive conditions. While both acids exhibit corrosiveness toward copper, there is a comparison of one to the other. The corrosion rate at 328 K in the blank nitric acid solution was 590.1 mpy while that in sulphuric acid was 147.8 mpy. The notable difference is due to the highly oxidizing nature of nitric acid causing copper to dissolve.

In both media, the inhibitor was very good, but with greater efficiency in sulphuric acid. The greater

adsorption and stability of the inhibitor film in H_2SO_4 results in improved protective effect.

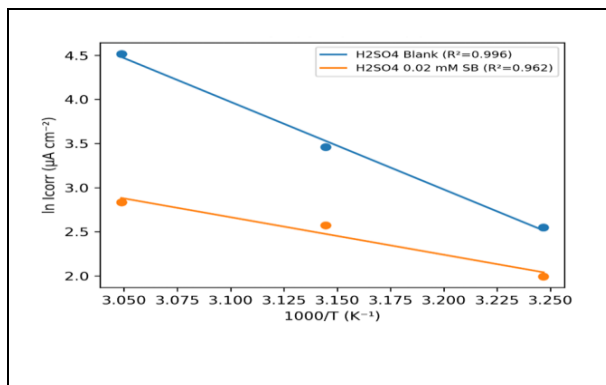


Fig.7a Arrhenius plots for the corrosion of copper in the absence and presence of N-(pyridin-2-ylmethylene) aniline in 1.0 M HNO_3 acid solution Top of Form

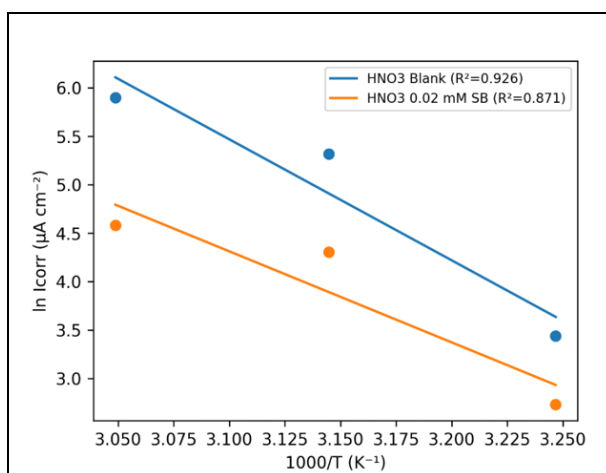


Fig.7b Arrhenius graphs for corrosion of copper in 0.5M H_2SO_4 acid solution in the absence and presence of N-(pyridin-2-ylmethylene) aniline

To further study the impact of temperature on the corrosion mechanism of copper in 1.0 M HNO_3 and 0.5 M H_2SO_4 solutions, Arrhenius and transition state analyses were applied. Table 5 shows the obtained thermodynamic parameters from measurements of polarization against temperature while the Arrhenius plots are shown in Fig. 7a and 7b. The activation process, adsorption behaviour and mechanism of inhibition of the Schiff base inhibitor can be deciphered from these parameters

The Arrhenius plots of $\ln(I_{\text{corr}})$ vs $1000/T$ show reasonably linear relationships indicating the Arrhenius behavior of the corrosion process in the temperature range studied. The values of the activation energy for dissolution of copper in the absence and presence of Schiff base inhibitor were determined from the slopes of the plots. The activation energy (E_a) of nitric acid solution (uninhibited) was found to be $103.79 \text{ kJ mol}^{-1}$. The

activation energy has been reduced at $78.09 \text{ kJ mol}^{-1}$ after the addition of 0.02 mM Schiff base.

The drop in activation energy with the addition of inhibitor suggests that the adsorption of the inhibitor on the copper surface becomes more favourable with temperature. Accordingly, such behavior is usually associated with chemisorptive interactions, wherein the inhibitor forms relatively strong bonds with the metal. The layer of molecules adsorbed on the metal surface modifies the corrosion pathway and lowers the energy barrier to corrosion.

The enthalpy of activation ($\Delta H^\ddagger$) of the blank solution was $101.15 \text{ kJ mol}^{-1}$ which decreased to $75.45 \text{ kJ mol}^{-1}$ for the inhibited solution. The fact that a positive value of $\Delta H^\ddagger$ indicates the dissolution of copper is an endothermic reaction, requiring energy for activation. The lower $\Delta H^\ddagger$ value of the inhibited system indicates that the adsorption of the inhibitor changes the surface environment.

The addition of inhibitor decreased the entropy of activation ($\Delta S^\ddagger$) from $113.45 \text{ J mol}^{-1} \text{ K}^{-1}$ to $24.18 \text{ J mol}^{-1} \text{ K}^{-1}$. It means that the disorder of activated complex is reduced. The organization of the interfacial structure due to the adsorbed inhibitor molecules results in a more organized state of the transition.

Table.5 Thermodynamic parameters of copper corrosion in 1.0M HNO_3 and 0.5M H_2SO_4 acid solutions in the absence and presence of Arrhenius plots explained the presence of N-(pyridin-2-ylmethylene) aniline in the solution.

System	E_a (kJ mol ⁻¹)	$\Delta H^\ddagger$ (kJ mol ⁻¹)	$\Delta S^\ddagger$ (J mol ⁻¹ K ⁻¹)
HNO_3 Blank	103.79	101.15	113.45
$\text{HNO}_3 + 0.02 \text{ mM SB}$	78.09	75.45	24.18
H_2SO_4 Blank	82.46	79.82	34.90
$\text{H}_2\text{SO}_4 + 0.02 \text{ mM SB}$	35.34	32.70	-122.03

In sulphuric acid medium, the activation energy for the blank solution was $82.46 \text{ kJ mol}^{-1}$ and this value is lower than that in nitric acid. The activation energy was further reduced to $35.34 \text{ kJ mol}^{-1}$ in the presence of 0.02 mM Schiff base.

The significant reduction of activation energy indicates a strong interaction between inhibitor molecules and the copper surface. The addition of the inhibitor effectively changes the electrochemical environment and produces a stable protective adsorbed film. The lower value of E_a in H_2SO_4 than HNO_3 indicates stronger adsorption and better inhibition performance of the inhibitor in sulphuric acid medium.

The enthalpy of activation for the blank solution was $79.82 \text{ kJ mol}^{-1}$ which decreased to $32.70 \text{ kJ mol}^{-1}$ for after addition of inhibitor. The positive value of $\Delta H^\ddagger$ confirms again that copper dissolution is endothermic (Frumkin et al. 2017). The lower enthalpy in the inhibited system indicates the inhibitor film stabilizes the copper surface (Villamil et al. 1999).

A significant change in the entropy of activation was noted. The $\Delta S^\ddagger$ value of the blank solution was determined to be $34.90 \text{ J mol}^{-1} \text{ K}^{-1}$ while the inhibited system showed a value of $-122.03 \text{ J mol}^{-1} \text{ K}^{-1}$. The negative entropy value suggests that a high order activated complex is formed during corrosion (Quraishi and Sardar et al. 2003; Ashassi-Sorkhabi and Shaabani et al. 2005). This behavior is consistent with the strong adsorption of inhibitor molecules and the formation of an organized surface film on copper.

The analysis of the thermodynamic parameters shows that Schiff base adsorbs more in sulphuric acid medium when compared to nitric acid medium. The addition of an inhibitor reduced the activation energy by about 25.7 kJ mol^{-1} in HNO_3 while it decreased by almost 47.1 kJ mol^{-1} in H_2SO_4 . In the same way, the entropy change became strongly negative in sulphuric acid indicating that a well-ordered adsorbed layer was formed.

These results correlate excellently with the data obtained from electrochemical impedance and polarization studies, which showed that the highest inhibition efficiencies in H_2SO_4 were superior to HNO_3 . The data related to thermodynamics suggest that adsorptions of Schiff base is more favorable and stable in sulfuric acid medium.

The lowered values of activation energy and enthalpy with considerable entropy changes, indicated that the process of inhibition occurs due to the adsorption of Schiff base molecules *via* nitrogen atom and conjugated π -electron system of benzene (Elachouri et al. 1996). The triple bond between nitrogen and copper is thus avoided, resulting in the formation of an adsorbed protective layer.

The results from the polarization studies indicated that the increase in inhibition efficiency with the rise in the temperature together with the decrease in activation energy suggested a considerable

chemisorption character of inhibition. The adsorbed layer effectively obstructs active corrosion positions and inhibits both anodic copper dissolution and cathodic reduction reaction.

3.7 Potential of Zero Charge

Differential capacitance measurements at copper were performed in view of assessing the interfacial behaviour in 1.0 M HNO_3 and 0.5 M H_2SO_4 solutions without and with the Schiff base inhibitor. Electrochemical parameters associated with surface charge are given in Table 6 and differential potential capacitance curves are seen in Fig. 8(a to d). These measures will provide useful information on the adsorption behavior of the inhibitor and the electrical properties of the copper/solution interface.

Fig. 8a shows the differential capacitance of copper in 1.0 M HNO_3 solution without inhibitor as potential changes. The lowest point on that curve is around -25 mV . This is the potential of zero charge (EPZC). When a potential is applied, the net charge on the metal surface becomes 0 and the electrostatic interaction between the electrode and electrolyte becomes minimum

As the applied potential shifts away from the E_{PZC} in the anodic or cathodic direction, the capacitance tends to increase gradually. This action is associated with alterations in the configuration of ions and water molecules within the electrical double layer (Obot and Obi-Egbedi et al. 2010). Increased capacitance values for increasing more positive as well as more negative potentials indicate enhanced accumulation of charged species at the metal-solution interface (El-Etre et al. 2003; Obot and Obi-Egbedi et al. 2010).

The open-circuit potential (E_{ocp}) for copper measured in 1.0 M HNO_3 was $+42.10 \text{ mV}$, while the E_{PZC} was -25.0 mV . The difference between the two values ($E_{ocp} - E_{PZC}$) is called excess charge and was calculated to be $+67.10 \text{ mV}$. A positive value indicates that the copper surface has a net positive charge in the corrosion condition.

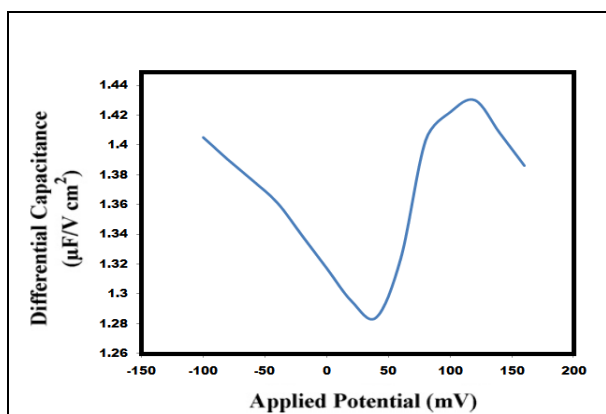


Fig.8a Differential capacitance versus Applied potential in 1.0 M HNO₃ acid solution

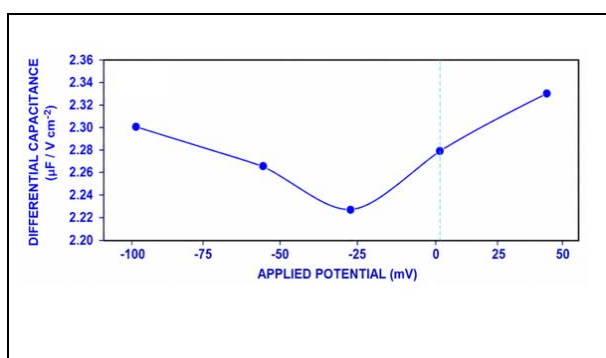


Fig.8b Differential capacitance versus Applied potential of copper in 1.0 M HNO₃ acid solution in the presence of N-(pyridin-2-ylmethylene) aniline

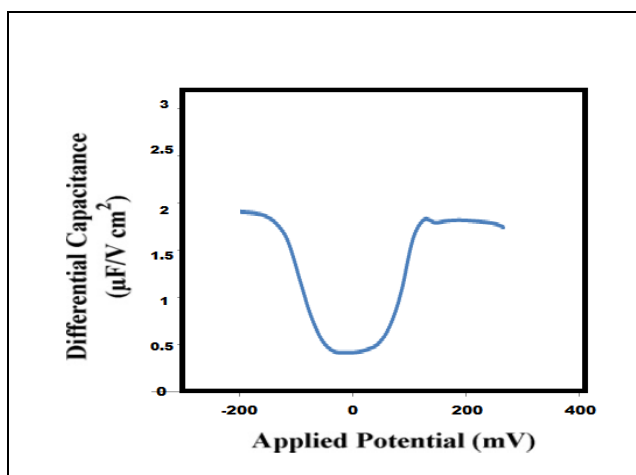


Fig.8c Differential capacitance vs Applied potential copper in 0.5 M H₂SO₄.

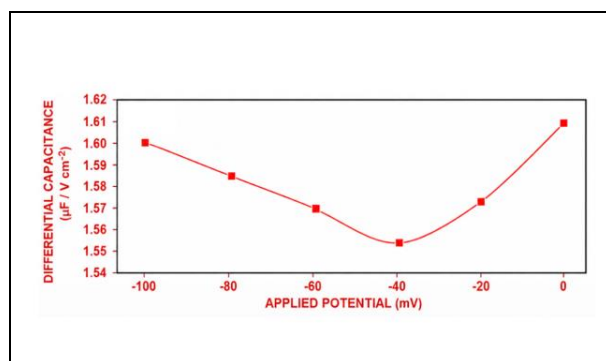


Fig.8d Differentiated capacitance vs Applied potential in copper solution in 0.5 M H₂SO₄ acid in the presence of N-(pyridin-2-ylmethylene) aniline

Table.6 Surface charge on the copper metal surface in 1.0M HNO₃ and 0.5M H₂SO₄ acid solutions in the presence and absence of N-(Pyridin-2-ylmethylene) aniline.

Medium	E _{OCP} (mV/SCE)	E _{PZC} (mV/SCE)	Excess Charge (mV)
1.0 M HNO ₃	+42.10	-25.0	+67.10
1.0 M HNO ₃ + 0.01 mM SB	-12.30	-25.0	+12.70
0.5 M H ₂ SO ₄	+4.52	-40.0	+44.52

As seen in Fig. 8b, the differential capacitance curve was obtained when 0.01 mM Schiff base inhibitor was added. The capacitance values are markedly changed compared with the uninhibited system, indicating adsorption of inhibitor molecules on the copper surface (Noor et al. 2005).

The minimum capacitance continues to be close to the potential of zero charge; the general shape of the curve becomes smoother and more symmetric. This type of behavior indicates the generation of a layer of adsorbed inhibitor onto the surface that modifies the structure of electric double layer.

The E_{OCP} changes from +42.10 mV to -12.30 mV after the addition of the inhibitor while E_{PZC} changes negligibly at -25.0 mV. As a result, the excess charge drops from 67.10 mV to 12.70 mV. This large depletion of the positive charge indicates that the inhibitor molecules neutralized the charge that is

present on the surface, which greatly reduces the electrostatic attraction between the metal surface and aggressive ions (Eddy and Ebenso *et al.* 2010; Verma *et al.* 2018; Saha *et al.* 2018).

The replacement of the water molecules by large organic inhibitor molecules may be responsible for the decrease in capacitance. As the dielectric constant of inhibitor layer is less and thicker than the water layer (i.e., original layer), thus capacitance is less (El-Etre *et al.* 2003; Obot and Obi-Egbedi *et al.* 2010). This observation gives strong evidence of adsorption of the Schiff base on to the copper surface.

The AC capacitance-potential curve for copper in 0.5M H₂SO₄ solution is shown in Fig. 8c. Like nitric acid system, capacitance curve shows well defined minimum that corresponds to zero charge potential. The curve shape shows the classical electrical double layer formation at the copper-electrolyte interface. The E_{OCP} value was obtained as +4.52 mV and the E_{PZC} was -40.0 mV. The excess charge obtained was +44.52 mV. The positive excess charge indicates that the copper surface in a sulphuric acid medium is positively charged in open circuit.

As may be seen, sulphuric acid has apparently higher capacitance values than nitric acid. This might have occurred due to differences in ionic adsorption properties and double-layer structure.

The capacitance-potential graph in the sulphuric acid medium of the Schiff base inhibitor was derived and shown in Fig. 8d. The values of capacitance decreased sharply after adding the inhibitor, indicating that inhibitor molecules got adsorbed at the metal surface. The rate of decline of the differential capacitance suggests that the adsorbed inhibitor molecules displace water molecules and increase the effective thickness of double layer. Simultaneously, organic films reduce the dielectric constant of the interface, yielding lower capacitance values.

The alterations observed indicate the formation of a dense and adherent protective layer, which seals the active corrosion sites on the copper surface. A barrier for both ion transport and electron transfer is provided, thus lower the corrosion rate.

The link between E_{OCP} and E_{PZC} has useful information relative to the type of adsorption. In acidic conditions, the E_{OCP} value remains more positive than the corresponding E_{PZC} value, resulting in positive excess charges. This suggests that the copper surface is positively charged in the corrosion situation.

In this context, the neutral inhibitor molecules can adsorb on the surface through donor-acceptor interaction involving the lone-pair electrons

of the nitrogen atom and π -electron cloud of the Schiff base. Moreover, negatively charged species which are absorbed from the electrolyte act as a bridge between the copper surface which is positively charged.

The resultant decrease in excess charge compared to blank after the addition of the inhibitor in nitric acid medium indicates that the adsorption of the Schiff base alters the range of charge at the interface.

The studies of differential capacitance provide evidence that the Schiff base inhibitor interacts strongly with the surface of copper. The decline in capacitance, the alteration in open-circuit potential, and the drop in excess charge all point to the formation of an adsorbed protective layer.

The nitrogen atom and conjugated aromatic system that is electron rich attaches themselves to the copper surface. The prepared layer limits the active corrosion sites, increases the metal-solution interface resistance and suppresses the electrochemical reactions that lead to the dissolution of copper.

The lowering of the differential capacitance is consistent with the earlier impedance and polarization investigations, whereby the charge-transfer resistance was raised and the corrosion current density lowered when the inhibitor was present.

3.8 Synergism Parameter

The impact of halide ions on the corrosion inhibition behavior of Schiff base inhibitor towards copper was examined through Electrochemical Impedance Spectroscopy (EIS) studies in 1.0 M HNO₃ and 0.5 M H₂SO₄ solutions. The Nyquist plot impedance parameters are shown in Table 7, while the impedance spectra are indicated in Figs 6.9a-6.9f. The analysis of the synergistic interaction of Schiff base inhibitors with various ions was carried out in the presence and absence of chloride, bromide and iodide ions.

The Nyquist plots obtained in acidic media display depressed capacitive semicircles, indicating that the charge-transfer process at the copper-solution interface is the rate-determining step for corrosion. As charge-transfer resistance (R_{ct}) increases, the diameter of the semicircle increases, which is directly related to corrosion protection efficiency.

The semicircle diameter had significantly increased when the Schiff base was only added when compared to the blank solution. This shows that there was already adsorption of inhibitor molecules on the copper surface (El-Etre *et al.* 2003; Obot and Obi-Egbedi *et al.* 2010). Introducing halide ions into the inhibited system led to further changes, suggesting that the inhibitor molecules interacted with the halide ions.

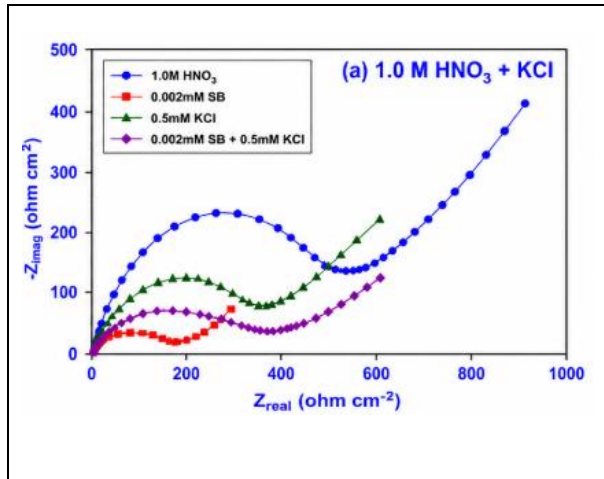


Fig.9a The Nyquist plots for Cu electrode obtained in 1.0 M HNO_3 solution in the absence and in the presence of SB and KCl

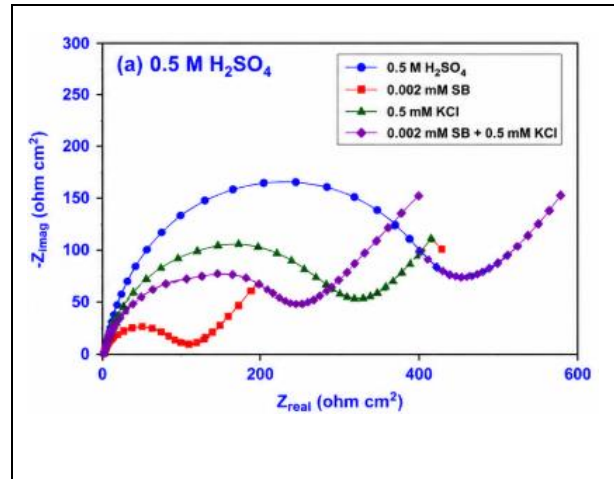


Fig.9d The Nyquist plots for Cu electrode obtained in 0.5 M H_2SO_4 solution in the absence and in the presence of SB and KCl

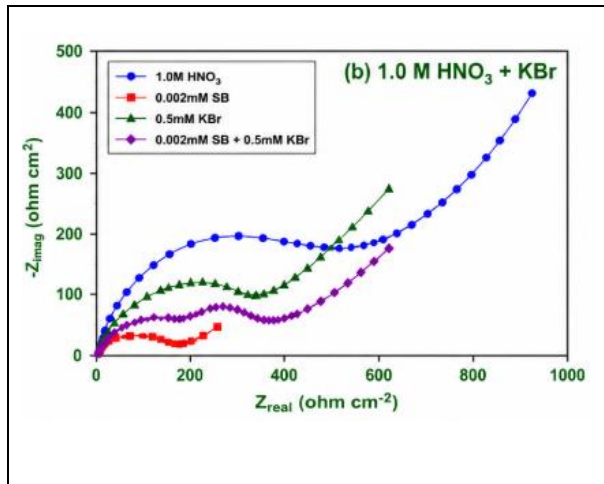


Fig.9b The Nyquist plots for Cu electrode obtained in 1.0 M HNO_3 solution in the absence and in the presence of SB and KBr

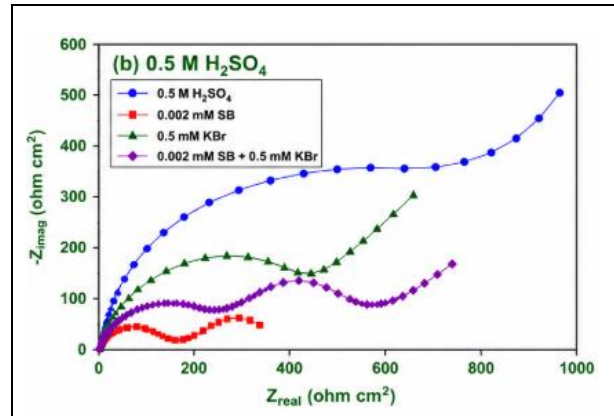


Fig.9e The Nyquist plots for Cu electrode obtained in 0.5 M H_2SO_4 solution in the absence and in the presence of SB and KBr

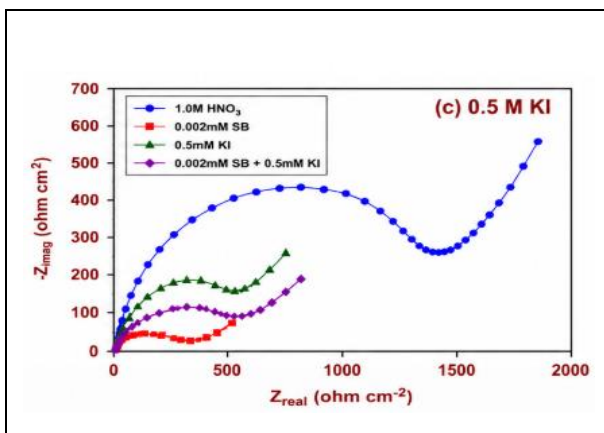


Fig.9c The Nyquist plots for Cu electrode obtained in 1.0 M HNO_3 solution in the absence and in the presence of SB and KI

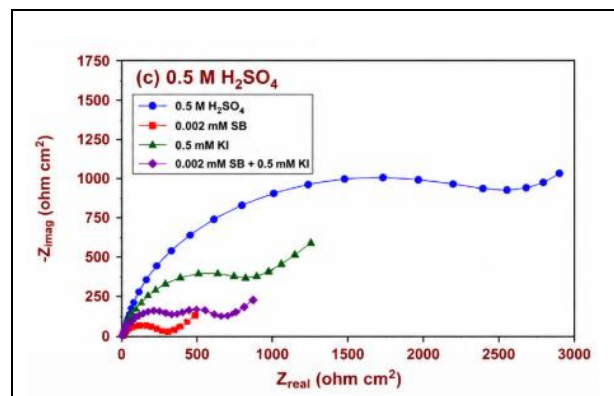


Fig.9f The Nyquist plots for Cu electrode obtained in 0.5 M H_2SO_4 solution in the absence and in the presence of SB and KI

The charge-transfer resistance of blank nitric acid solution was only $89.0 \Omega \text{ cm}^2$ showing that copper may corrode quickly. The Schiff base at the concentration of 0.002 mM raised R_{ct} to $195.8 \Omega \text{ cm}^2$, indicating the protective effect of the inhibitor.

The addition of 0.5 mM KCl alone caused the R_{ct} value to increase to $318.6 \Omega \text{ cm}^2$ indicating surface adsorption due to chloride ions. The presence of Schiff base and KCl further increase the R_{ct} to $502.4 \Omega \text{ cm}^2$ and $\theta = 0.8273$.

However, the value of the synergism parameter (S_θ) was found to be 0.7351 . A value of S_θ lesser than unity indicates antagonistic behaviour, signifying that chloride ions partially compete with the inhibitor molecules at the copper adsorption sites. As a result, the total inhibition effect is lower than that from ideal cooperative adsorption.

The addition of KBr alone increased R_{ct} to $236.4 \Omega \text{ cm}^2$, whereas R_{ct} for combined system containing Schiff base and KBr was $462.8 \Omega \text{ cm}^2$ with $\theta = 0.8118$.

The value of synergism parameter was 0.9024 (less than unity). This indicates that the bromide ions and Schiff base interact at a weak level. Even though bromide ions could promote corrosion resistance, the extent of this improvement is not as considerable as the real synergistic adsorption would require to bring about.

The behavior of iodide ion is quite different from those of chloride and bromide ions. The result of combination of Schiff base with KI gives very high value of R_{ct} that is $965.3 \Omega \text{ cm}^2$. This value is much higher than that obtained using KCl and KBr.

The findings revealed that surface coverage of 0.9124 and a synergism parameter of 3.47 , which is much greater than unity. The high S_θ value confirms the strong synergistic effect of iodide ions on the Schiff base inhibitor (Olasunkanmi *et al.* 2021).

Iodide ions perform well because they are large, highly polarizable and possess a strong affinity to attach to the copper surface (Avdeev *et al.* 2018; Roscher *et al.* 2024; Fiala *et al.* 2007). The adsorbed iodide ions form negative sites on the surface. This allows for the adsorbed

protonated Schiff base molecules to effectively inhibit corrosion.

A comparable tendency was noted in sulphuric acid experiments. The R_{ct} value of the blank solution was found to be $65.1 \Omega \text{ cm}^2$, while the Schiff Base alone increased the R_{ct} to $392.5 \Omega \text{ cm}^2$ which corresponds to a surface coverage of 0.7783 .

The R_{ct} (charge transfer resistance) value of Schiff base in the presence of KCl was $253.7 \Omega \text{ cm}^2$ with $\theta = 0.7498$.

The synergy parameter was 0.3056 , indicating a strong antagonistic interaction. Thus, the adsorption of Schiff base molecules is affected by chloride ions and there is a decrease in overall protection.

When KBr was added to the combination inhibitor system, the R_{ct} was $614.5 \Omega \text{ cm}^2$ and the surface coverage value was 0.8987 . The value of S_θ attained was 0.8912 , which is slightly less than unity. Even though the bromide ion is a better inhibitor than the chloride ion, it doesn't interact with the Schiff base.

Here, KI showed the most considerable improvement in the outcome. The combination of the Schiff base and iodide systems showed an R_{ct} value of $1096.8 \Omega \text{ cm}^2$ which was the highest value among all the systems tested.

As the parameter of surface coverage becomes 0.9425 and the synergism parameter equals 3.58 , this is strong cooperation between the iodide ions and those of the inhibitor. The enlarged semicircle in Fig 6.9f shows that a highly resistive protective layer prevents electron transfer and copper dissolution.

The synergism parameter (S_θ) gives quantitative information about the interaction of inhibitor molecules and halide ions.

$S_\theta > 1$ indicates synergistic behaviour.

$S_\theta = 1$ indicates independent adsorption.

$S_\theta < 1$ indicates antagonistic behaviour.

Table.7 The effect of halides ion on copper corrosion of 1.0 M HNO_3 and $0.5 \text{ M H}_2\text{SO}_4$ acid solution in the absence and presence of N-(pyridin-2-ylmethylene)aniline measured using Electrochemical impedance spectroscopy.

Medium	Inhibitor Concentration (mM)	$Y_0 (\times 10^{-6} \Omega^{-1} \text{ cm}^{-2})$	n	$R_{ct} (\Omega \text{ cm}^2)$	θ	S_θ
1.0 M HNO_3	Blank	856.4	0.996	89.0	–	–
	0.002 mM SB	196.3	0.924	195.8	0.5454	–

Medium	Inhibitor Concentration (mM)	$Y_0 (\times 10^{-6} \Omega^{-1} \text{ cm}^{-2})$	n	$R_{ct} (\Omega \text{ cm}^2)$	θ	S_0
0.5 M H ₂ SO ₄	0.5 mM KCl	185.7	0.901	318.6	0.7206	–
	0.002 mM SB + 0.5 mM KCl	55.2	0.972	502.4	0.8273	0.7351
	0.5 mM KBr	267.8	0.914	236.4	0.6235	–
	0.002 mM SB + 0.5 mM KBr	137.5	0.965	462.8	0.8118	0.9024
	0.5 mM KI	101.8	0.907	126.7	0.2976	–
	0.002 mM SB + 0.5 mM KI	108.6	0.927	965.3	0.9124	3.47
	Blank	326.8	0.932	65.1	–	–
	0.002 mM SB	199.4	0.923	392.5	0.7783	–
	0.5 mM KCl	344.7	0.946	196.3	0.6684	–
	0.002 mM SB + 0.5 mM KCl	354.8	0.944	253.7	0.7498	0.3056
	0.5 mM KBr	252.9	0.936	158.2	0.5889	–
	0.002 mM SB + 0.5 mM KBr	149.8	0.962	614.5	0.8987	0.8912
	0.5 mM KI	148.3	0.928	178.6	0.6358	–
	0.002 mM SB + 0.5 mM KI	118.6	0.944	1096.8	0.9425	3.58

Thus, chloride and bromide ions exhibit antagonistic behavior in both acidic media, whereas iodide ions display a pronounced synergistic effect.

The greater inhibition of iodide ions may arise from the increased adsorption property of iodide ions. The larger and more polarizable nature of iodide ions renders them superior adsorbates onto the copper surface than both chloride and bromide ions (Chen et al. 2024; Yuan et al. 2023). Once adsorbed, the iodide ions generate negatively charged sites that favor electrostatic attraction of protonated Schiff base molecules. The being coherent while absorbing material increases coverage and helps in the formation of a thick and stable coating. In comparison to bromide, chloride shows a weaker adsorption, which means that they may also compete with the adsorbing inhibitor molecules for the active sites and thereby reduce the effectiveness of the adsorbed inhibitor layer.

3.9 Quantum Chemical Calculations

One of the most effective methods used to reduce copper corrosion is through the adsorption of organic molecules with heteroatoms and π -electron systems. The quantum chemical parameters of the synthesized Schiff base N-(pyridin-2-ylmethylene)aniline and of its protonated forms were

found to be responsible for the corrosion inhibition behavior. The optimized molecular structures shown in Fig 6.10 indicate that the inhibitor has two nitrogen atoms and conjugated aromatic rings that may serve as active adsorption sites. These electron-rich site facilitates interaction between inhibitor molecules and copper surface *via* donor-acceptor mechanism. The shape of the molecule also helps to surface coverage on the metal interface.

Based on HOMO distributions, the highest occupied molecular orbital is mainly confined on the aromatic rings and N-containing groups. The molecule will be more prone to donate electrons to the empty orbitals of copper atoms if the HOMO energy is higher. The protonated species have higher HOMO energies compared to the neutral molecule, implying stronger electron-donating ability and enhanced adsorption onto the copper surface. It makes the active corrosion sites less available and conceals the anodic metal dissolution.

The LUMO distributions illustrate that the inhibitor molecules can receive electrons from the filled d-orbitals of copper. The strengthening of the interaction between inhibitor and metal occurs due to this back-donation process. The characterization of Schiff base showed the presence of electron-donating and electron-accepting character. This shows that the copper can bond

with the Schiff base through synergistic adsorption having coordinate bonding and electrostatic attraction.

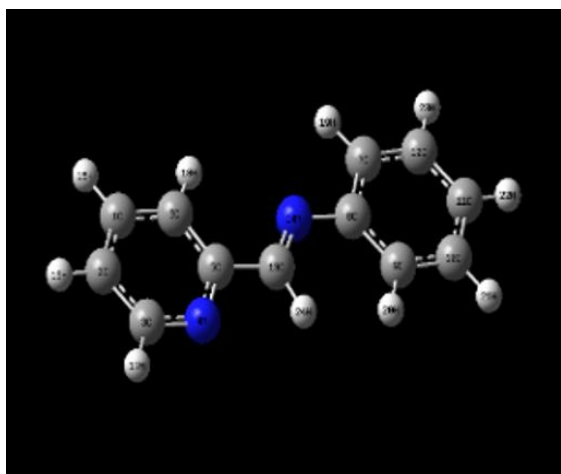


Fig.10a Optimized structure of synthesized Schiff base N-(pyridin-2-ylmethylene) aniline

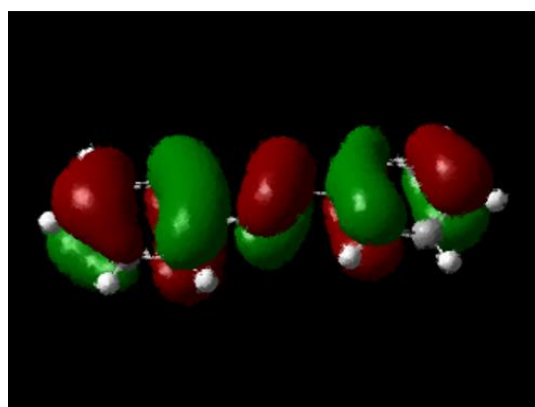


Fig.10b HOMO of N-(pyridin-2-ylmethylene) aniline

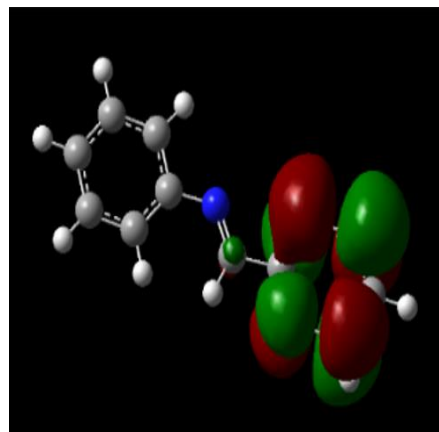


Fig.10c LUMO of N-(pyridin-2-ylmethylene) aniline

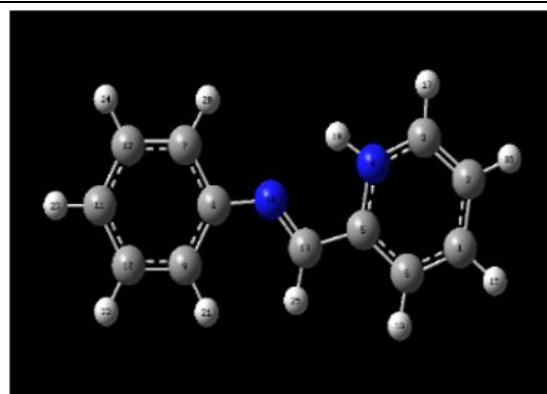


Fig.10d Optimized structure of N-(pyridin-2-ylmethylene) aniline protonated at 4N

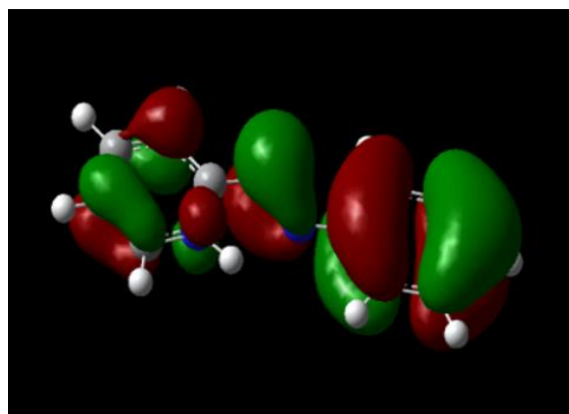


Fig.10e HOMO of N-(pyridin-2-ylmethylene) aniline protonated at 4N

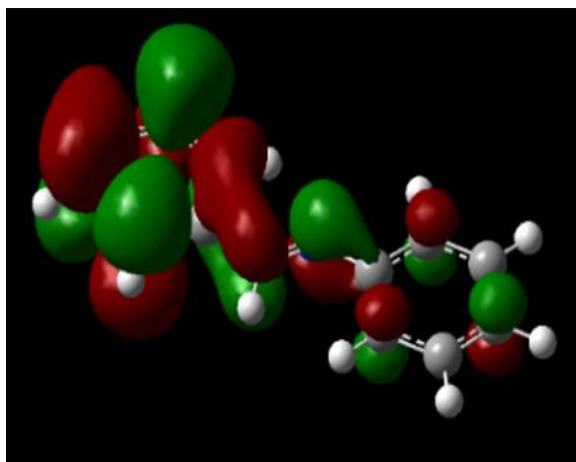


Fig.10f LUMO of N-(pyridin-2-ylmethylene) aniline protonated at 4N

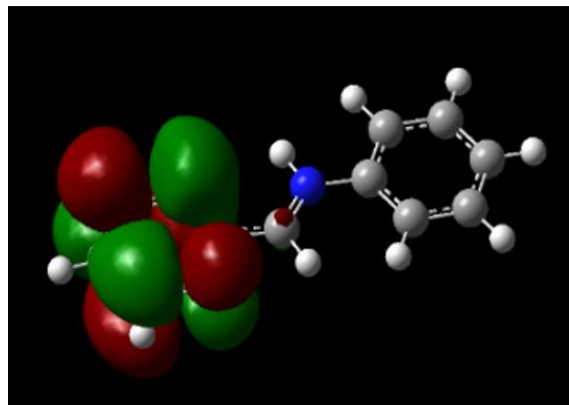


Fig.10i: LUMO of N-(pyridin-2-ylmethylene) aniline protonated at 14N

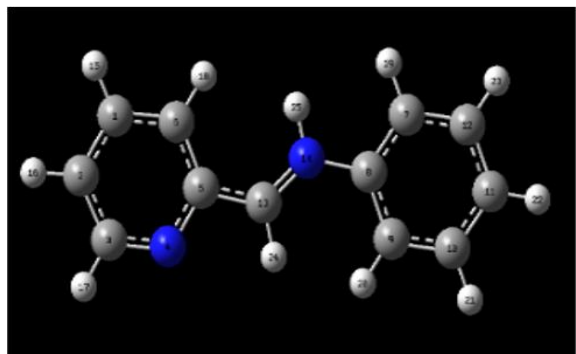


Fig.10g Optimized structure of N-(pyridin-2-ylmethylene) aniline protonated at 14N

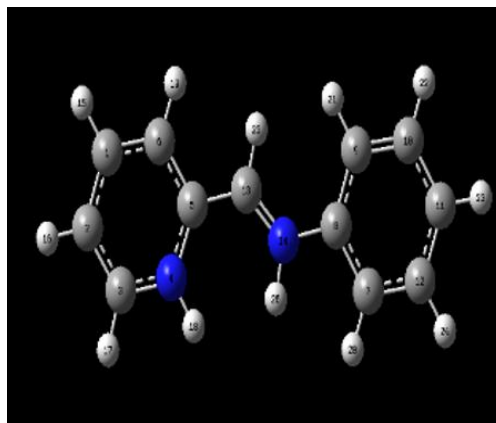


Fig.10j: Optimized structure of N-(pyridin-2-ylmethylene) aniline protonated at 4N and 14N

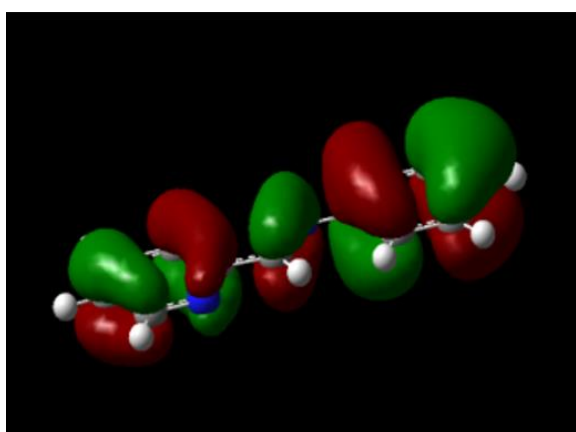


Fig.10h: HOMO of N-(pyridin-2-ylmethylene) aniline protonated at 14N

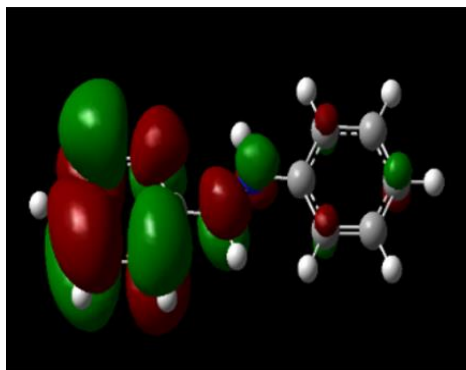
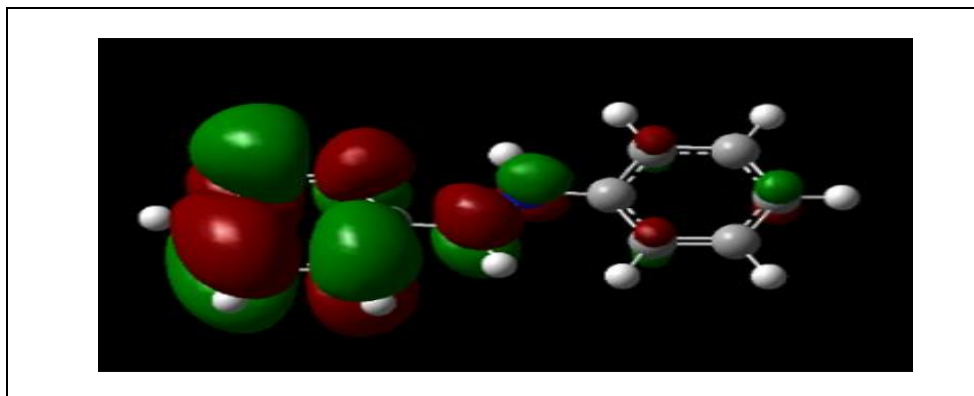


Fig.10k: HOMO of N-(pyridin-2-ylmethylene)aniline protonated at 4N and 14N**Fig.10l:** LUMO of N-(pyridin-2-ylmethylene) aniline protonated at 4N and 14N**Table.8** Quantum chemical Parameters obtained in the N-(pyridin-2-ylmethylene) aniline and the protonated forms

S. No.	QC Parameters	Neutral	Protonated at 4N	Protonated at 14N	Protonated at 4N & 14N
1	Total Energy (eV)	-1981.426	-1995.184	-11372.865	-2008.743
2	EHOMO (eV)	-9.218	-7.672	-7.204	-7.068
3	ELUMO (eV)	-1.064	-0.039	-0.322	-0.043
4	ΔE (eV)	8.154	7.633	6.882	7.025
5	Ionization Potential, I (eV)	9.218	7.672	7.204	7.068
6	Electron Affinity, A (eV)	1.064	0.039	0.322	0.043
7	Hardness, η (eV)	4.077	3.817	3.441	3.513
8	Softness, S (eV^{-1})	0.123	0.131	0.146	0.143
9	Electronegativity, χ (eV)	5.141	3.856	3.763	3.556
10	Electrophilicity Index, ω	3.242	1.948	2.057	1.798
11	Fraction of Electrons Transferred, ΔN	0.231	0.414	0.472	0.493

The energy gap (ΔE) between HOMO and LUMO can indicate the reactivity of the molecular. The neutral

molecule has higher energy gap values than that of the protonated species. A lower ΔE means the molecule is

softer and has a greater tendency to adsorb, allowing the inhibitor molecules to interact more easily with the copper surface (Madkour and Elshamy *et al.* 2016). As a result, it is expected that the protonated forms will have a better efficiency than quinoidal forms for corrosion inhibition in acidic media.

The calculated hardness (η) and softness (S) scores support this observation further. Low hardness and high softness indicate high molecular polarizability and also have the capability to exchange electrons with the metal. The investigated species all show that the softness of their protonated forms could mean stronger adsorption and better corrosion protection (Madkour and Elshamy *et al.* 2016). This allows the formation of a compact barrier film which restricts the transport of the corrosive ionic species to copper.

The fraction of electrons exchanged significantly increases upon protonation. Higher ΔN values suggest that the inhibitor is more likely to donate electrons to copper, indicating stronger chemisorption and increased stability of the adsorbed film. The ΔN value for the protonated molecule at both nitrogen sites is maximum which indicates strongest interaction with the metal surface least inhibition expected. According to the values of electrophilicity index and electronegativity, the protonation modifies the electronic structure of the Schiff base, thus increasing the adsorption ability (Madkour and Elshamy *et al.* 2016). The balance between donating and accepting electrons helps achieve the optimum surface interaction and leads to the formation of a stable protective layer on copper (Zhang *et al.* 2021; Cao *et al.* 2025). This adsorbed layer acts as a physical and chemical barrier to anodic copper dissolution and cathodic reduction. Based on quantum chemical analysis the N-(pyridin-2-ylmethylene) aniline can be a good corrosion inhibitor for copper.

The presence of nitrogen atoms, conjugated π -electron systems, advantageous HOMO–LUMO properties, smaller energy gap values, increased softness, and higher electron transfer capacities together support adsorption on the copper surface. The protonated forms show better electronic properties than the neutral counterpart which suggest stronger metal–inhibitor interactions and better inhibition. Hence, the mechanism of corrosion protection is mainly governed by the adsorption on the copper surface of the Schiff base molecules and consequently the formation of a remarkably stable and adherent protective film that efficiently retards copper corrosion.

4. CONCLUSION

This research shows the synthesized Schiff base, N-(pyridin-2-ylmethylene) aniline, is an effective corrosion inhibitor for copper in 1.0 M HNO_3 and 0.5 M H_2SO_4 solutions. The copper surface was successfully adsorbed onto the Schiff base in detail spectroscopic characterization attributed to the presence of the characteristic azomethine ($-\text{C}=\text{N}-$) functional group.

Results of weight-loss measurements indicated that the inhibition efficiency increased with increasing inhibitor concentration in both acidic media. This result showed that the inhibitor molecules get adsorbed on copper surface in a concentration-dependent manner. Maximum efficiency of protection was obtained at high concentration of inhibitors, and performance was superior in sulphuric acid medium. This enhanced protection is due to the compact and adherent protective film that effectively isolates the metal from corrosive environment.

Studies using electrochemical impedance spectroscopy confirmed that the Schiff base is protective. The rise in charge-transfer resistance with a fall in double-layer capacitance indicated the development of insulated adsorbed layer on copper surface. The corrosion rate was lowered by the inhibitor by effectively restricting the charge transfer and successively hindering aggressive ions' diffusion. The larger Nyquist semicircles in the presence of inhibitor indicated greater surface coverage and stronger adsorption.

Studies of potentiodynamic polarization reveal that the inhibition is both anodic (copper dissolution) and cathodic (oxygen reduction) in nature. The decrease in corrosion current density was large, indicating the effectiveness of the adsorbed film controlling the corrosion-electrochemical process as the concentration of the inhibitor increases. The relatively minor changes in corrosion potential supported the mixed-mode inhibition mechanism.

Adsorption studies exhibited good fit to Langmuir adsorption isotherm indicating the formation of monolayer of inhibitor on copper surface. The free energy of adsorption values being negative shows that the adsorption process is spontaneous and thermodynamically favorable. The adsorption equilibrium constant of the inhibitor increases and the more negative free energy ($\Delta G_{\text{ads}}^\circ$) values obtained in sulphuric acid medium show a stronger interaction of the inhibitor with the copper surface. Thus, more protection of copper corrosion is observed.

Studies conducted at different temperatures indicate that the inhibitor is effective over a wide range of temperatures. As the temperature rises, the effectiveness of inhibiting improves. It indicates an increase in the interaction of inhibitor molecules with the

copper surface, which created a more stable and compact layer. This behavior indicates that Schiff base is adequate for practical applications at different operating conditions.

The strong corrosion inhibition performance of N-(pyridin-2-ylmethylene) aniline is due to the electron-rich nitrogen atoms and conjugated π -electron systems allowing it to bond effectively on the copper surface. The establishment of a consistently protective layer successfully impedes active corrosion sites and curtails metal dissolution. According to the cumulative outcome of gravimetric, electrochemical, adsorption and thermodynamic studies, this Schiff base can be used as an effective and eco-friendly inhibitor for the protection of copper in acidic media especially in sulphuric acid.

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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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